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The inherence heuristic: An intuitive means of making sense of the world, and a potential precursor to psychological essentialism

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Abstract: We propose that human reasoning relies on an *inherence heuristic*, an implicit cognitive process that leads people to explain observed patterns (e.g., girls wear pink) predominantly in terms of the inherent features of their constituents (e.g., pink is a delicate color). We then demonstrate how this proposed heuristic can provide a unified account for a broad set of findings spanning areas of research that might at first appear unrelated (e.g., system justification, nominal realism, is–ought errors in moral reasoning). By revealing the deep commonalities among the diverse phenomena that fall under its scope, our account is able to generate new insights into these phenomena, as well as new empirical predictions. A second main goal of this article, aside from introducing the inherence heuristic, is to articulate the proposal that the heuristic serves as a foundation for the development of psychological essentialism. More specifically, we propose that *essentialism* – which is the common belief that natural and social categories are underlain by hidden, causally powerful essences – emerges over the first few years of life as an elaboration of the earlier, and more open-ended, intuitions supplied by the inherence heuristic. In the final part of the report, we distinguish our proposal from competing accounts (e.g., Strevens's K-laws) and clarify the relationship between the inherence heuristic and related cognitive tendencies (e.g., the correspondence bias). In sum, this article illuminates a basic cognitive process that emerges early in life and is likely to have profound effects on many aspects of human psychology.

Keywords: correspondence bias; development; essentialism; explanation; inherence heuristic; is–ought problem; nominal realism; system justification

1. Introduction

The ability to identify and exploit the predictable aspects of a complex environment is, without doubt, part of what makes humans such a successful species (e.g., Murphy 2004; Saffran et al. 1996; Zhao et al. 2013). Even the youngest members of our species are able to detect the broad patterns that characterize their world: that boys wear blue and girls wear pink, that orange juice is consumed for breakfast, that giraffes are called *giraffes*, and so on. In the present article, we propose that people often make sense of such regularities via a simple rule of thumb – the *inherence heuristic*. This fast, intuitive heuristic leads people to explain many observed patterns[1] in terms of the inherent features of the things that instantiate these patterns. For example, one might infer that girls wear pink because pink is a delicate, inherently feminine color, or that orange juice is consumed for breakfast because its inherent qualities make it suitable for that time of day. As is the case with the output of any heuristic, such inferences can be – and often are – mistaken. Many of the patterns that currently structure our world are the products of complex chains of historical causes rather than being

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simply a function of the inherent features of the entities involved. The human mind, however, may be prone to ignore this possibility. If the present proposal is correct, people often understand the regularities in their environments as inevitable reflections of the true nature of the world rather than as end points of event chains whose outcomes could have been different.

Consider the color/gender mapping example. Although pink and blue are now unmistakably gendered, during the nineteenth and early twentieth centuries those colors were actually viewed as interchangeable "nursery colors" that symbolized the young age of the children who wore them, not their gender (Paoletti 2012, Ch. 5). What's more, on the occasions when particular colors were suggested for one gender or the other, the recommended mapping was often the *opposite* of what it is today. For instance, in the November 1890 issue of the *Ladies' Home Journal*, readers are advised in no uncertain terms that they should use "blue for girls and pink for boys, when a color is wished" (Hooper 1890). Therefore, the current color/gender mapping (pink for girls, blue for boys) is best explained not by the inherent perceptual properties of pink and blue but rather by the confluence of now-forgotten historical developments (e.g., marketing campaigns by department stores and clothing manufacturers [Paoletti 2012]). And yet, for many people living today, the pink/girl and blue/boy mappings feel natural and inevitable, as if there were something inherently feminine about pink and inherently masculine about blue (e.g., Hurlbert & Ling 2007). The thought of dressing their boys in an all-pink outfit, let alone a pink dress, would make many parents feel uncomfortable and would probably draw vehement protests from the children themselves – at least from those old enough to have detected the relevant regularity (e.g., LoBue & DeLoache 2011).

A similar argument applies to the case of orange juice being consumed for breakfast. Contrary to what our inherence-based intuitions may lead us to believe, the fact that we currently drink OJ for breakfast is largely a matter of historical accident – in particular, an extensive marketing campaign by the California Fruit Growers Exchange, which was saddled with a consistent glut of citrus fruit in the early 1900s and was looking for new ways to market its product (Laszlo 2007, Ch. 7). To illustrate, the *Annual Report of the General Manager of the California Fruit Growers Exchange* for the year 1929 boasts a magazine marketing campaign "with a total of 310,964,842 impressions" that "reached 28½ million homes" and featured, among other things, "orange juice for breakfast" (p. 18). The campaign appears to have been an instant and smashing success, enabling orange growers to sell even portions of their crop that were "formerly considered undesirable." Thus, orange juice went from being a novelty drink to a breakfast staple largely because of concerted efforts by orange producers to make it so, and not because its inherent properties made it an obvious choice for breakfast.

If, as hypothesized, people explain observed patterns mostly in terms of the inherent features of their constituents, this perspective cannot be an automatic consequence of the brute statistical facts. For example, the mere existence of a pattern whereby girls wear pink is not, in and of itself, informative about the reason for such a pattern. In principle, thinking that girls wear pink for reasons

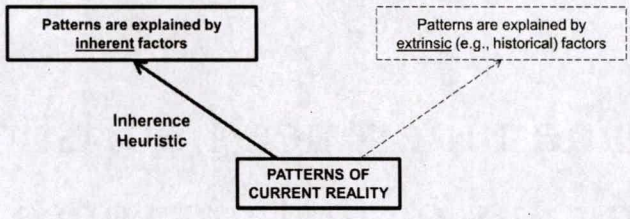


Figure 1. Schematic representation of the inherence heuristic.

extrinsic to both girls and pink (e.g., it's just a convention) would be as legitimate as thinking that girls wear pink because of something *inherent* to pink or girls (e.g., pink is a delicate color; girls have a "hardwired" attraction to pink). Therefore, a consistent preference for thinking that observed patterns are explained by inherent, rather than extrinsic, factors may speak to the rules of thumb that guide how people make sense of the world (see Fig. 1). More specifically, such a preference may speak to the operation of the hypothesized inherence heuristic, which leads people to interpret many broad facts about the world as being the by-products of inherent factors. In this article, we will argue that this heuristic is a pervasive feature of human cognition. At some point or another, humans have reasoned inherently about all sorts of patterns that actually arose from mutable historical forces (e.g., caste systems, child labor, women's confinement to the home) – just as people today tend to explain many of the regularities that structure their lives as the by-products of inherent features (e.g., Jost & Banaji 1994).

The rest of this article is structured as follows. In section 2, we explain how the hypothesized inherence heuristic may work: for example, what sort of mental process it is, what its inputs and outputs are, what other inferences it licenses, and how it may be overcome. In section 3, we argue that the inherence heuristic can provide a unified explanation for a number of disparate psychological phenomena. Then, in section 4, we propose that the inherence heuristic is a necessary ingredient in the process by which humans construct beliefs about the existence of physical, internal essences that define and explain how the natural and social worlds are carved up into kinds (e.g., Gelman 2003; Haslam et al. 2000; Medin & Ortony 1989). Our proposal that the inherence heuristic lays the foundation on which these so-called *essentialist* beliefs are constructed may shed new light on their origins, which are currently something of a mystery. Finally, section 5 clarifies the relationship between our proposal and other hypotheses that seek to account for some of the same phenomena (Prasada & Dillingham 2006; 2009; Strevens 2000), as well as related cognitive biases (e.g., the correspondence bias).

2. The inherence heuristic: What is it, and how does it work?

This section spells out how the inherence heuristic is hypothesized to operate. We begin by clarifying our use of the term *heuristic*, whose multiple senses may otherwise obscure what sort of cognitive process we have in mind.

2.1. Two types of heuristics: Deliberate and intuitive[2]

The term *heuristic* applies to two distinct classes of mental processes (e.g., Evans 2009; Frederick 2002; Gilovich & Griffin 2002). Some heuristics are deliberate strategies or procedures that we use to simplify complex problems. For example, when looking to buy a new car, one could decide to avoid the trouble of visiting multiple dealerships and to instead purchase the car that has the best online reviews within one's price range. Such voluntary searches for simple solutions are termed *deliberate* heuristics and can be contrasted with heuristics that operate at a more implicit level – *intuitive* heuristics. For example, the choice of a new car may also be swayed by impressions that pop into one's mind spontaneously, without any apparent effort (e.g., a hybrid would be nice). These easy intuitions also help narrow down answers to problems that would otherwise be complex and time-consuming, just as deliberate heuristics do; however, the two operate via different processes. Rather than being the output of a conscious decision-making process aimed at saving effort, the answers suggested by intuitive heuristics are the result of fast implicit processes that are automatically triggered by the problem under consideration (e.g., Frederick 2002; Kahneman 2011).

We now go on to explain the process by which the inherence heuristic is proposed to operate. In doing so, we draw on the compelling account synthesized by Kahneman (2011) out of four decades of empirical research on intuitive heuristics.

2.2. How does the inherence heuristic work?

The process underlying the inherence heuristic is set in motion when people seek to explain observed patterns (see Fig. 2). One of the fundamental conclusions of modern psychology is that humans have a powerful drive to make sense of their environments, a drive that prompts them to seek explanations spontaneously, on a routine basis, and from the earliest ages (e.g., Gopnik 1998; Gopnik et al. 2004; Lipton 2004; Murphy & Medin 1985; Premack & Premack 1996; Ross 1977; Weiner 1985). Even infants seem to posit unseen causal mechanisms to explain the evidence gathered from their interactions with the world (e.g., Gweon & Schulz 2011; Saxe et al. 2005; Schulz 2012). Although infants (and laypeople in general) may not approach the task of generating explanations in a terribly systematic and rigorous manner, they nevertheless show a deep-seated motivation to uncover the underlying structure of reality.

Once this explanatory drive is targeted at a particular pattern (e.g., Why do we drink orange juice for breakfast? Why do girls wear pink?), the next stage of the heuristic process is activated (see Fig. 2). Adapting one of Kahneman's (2011) terms, we call this stage the *mental shotgun*: the process of quickly activating any easily accessible information that might be relevant to answering the question at hand (see also Evans 2006; Stanovich 1999; 2011).[3,4] In the case of the inherence heuristic, then, the mental shotgun stage is likely to consist of a fast, shallow search for information that might be applicable (Higgins 1996) to the task of constructing an explanation for the pattern under consideration. In the rare cases when a specific answer is already known, the process terminates here. However, under most circumstances, the shotgun search will culminate not in the retrieval of a stored answer but rather in the generation of an assortment of facts that are potentially relevant to finding an answer.

Although the content generated by the mental shotgun will undoubtedly vary depending on the pattern to be explained, this content may nevertheless be structured along predictable lines. Because the shotgun prioritizes speed and ease of access, on most occasions it will start its search with the entities that happen to be most prominent in our minds at the point when the heuristic process is triggered. In the case of the inherence heuristic, these entities are typically the constituents of the pattern we are trying to explain. For instance, because people are already thinking about OJ and breakfast when they start wondering what explains their pairing, the mental shotgun will probably target those focal objects first. To be more specific, the shotgun is likely to activate any information that it has easy access to regarding these focal objects. What information might this be? Given that an object's representation in semantic memory often consists of information about its stable, inherent characteristics (e.g., McRae & Jones 2013), we expect that the output of the shotgun will correspondingly be dominated by the stable, inherent features of the participants in the relevant pattern (e.g., OJ smells refreshing; breakfast is in the

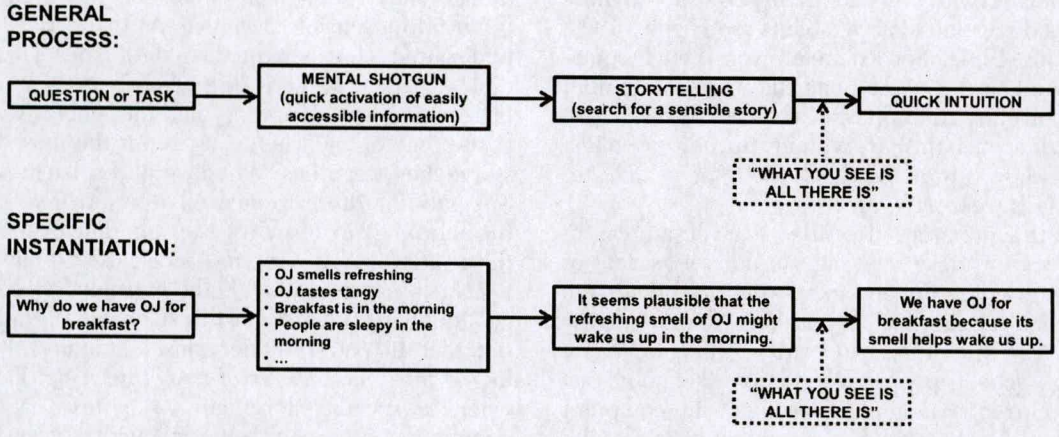


Figure 2. The general process involved in generating an intuitive judgment (*top*), and a specific instantiation of this process that leads to an inherence-based explanatory intuition (*bottom*).

morning). Semantic-associative information of this sort is highly accessible to implicit cognitive processes (e.g., Devine 1989; Greenwald et al. 1998; McRae et al. 1997; Rosch & Mervis 1975) and has in fact been implicated in the operation of other intuitive heuristics (e.g., Gilovich et al. 2002; Kahneman 2011; Sloman 1996).

As a side note, this description of the mental shotgun is compatible with the well-established principles that govern the process of knowledge activation, and in particular with Higgins's (1996) evidence of a "salience–applicability–knowledge activation" chain: "Salience...can impact subsequent responses by influencing which features of a stimulus event receive attention, and this in turn will influence which stored knowledge units are likely to be activated in the immediate situation" (p. 158). Because the focal entities of a pattern are typically the most salient at the point when the explanation question is posed, these entities will typically be deemed by the shotgun to provide information that is applicable (or relevant) to its task. This judged applicability will then prompt the shotgun to activate further knowledge about the focal entities – especially easily accessible knowledge about their inherent characteristics, which are tightly bound up with the representation of these entities in semantic memory.

As may already be apparent, we expect that the output of the mental shotgun will typically *fail* to include much information about past circumstances or external events pertinent to the pattern under consideration. One obvious reason for this failure is that such information (about past marketing campaigns, historical events, etc.) might be unknown to most people. However, even if such information were available, it might still not be picked up by the mental shotgun because this information is typically neither salient nor accessible. Unlike the constituents of the pattern to be explained, which loom large at the time when the inherence heuristic is triggered, the past circumstances that may have contributed to this pattern are often no longer in place and may also have no obvious physical connection with the pattern itself (for similar arguments, see Gilbert & Malone 1995). The inconspicuousness of these extrinsic factors makes it likely that they will be overlooked by the shotgun, even if knowledge about them was available. A second, related reason why extrinsic factors may not make it into the output of the mental shotgun is that information about them is often not as *accessible* as the information about inherent features, which is activated and consolidated with every additional exposure to these features (for a discussion of accessibility, see Higgins 1996; Higgins & King 1981). For example, even though somebody may have heard at some point that the color/gender mapping is currently the opposite of what it used to be, this piece of information may, without further consolidation, fail to show up on a quick shotgun search for reasons why girls wear pink.

The heuristics literature describes many examples of similar failures to retrieve relevant, but not very salient or accessible, knowledge. For instance, when told that Tom W. was "of high intelligence" but lacked "true creativity" and had "a need for order and clarity" when he was a senior in high school, people ranked the likelihood that Tom W. is currently a graduate student in computer science much higher than the likelihood that he is a graduate student in the social sciences (Kahneman & Tversky 1973). This typical, intuitive answer overlooks a crucial piece of information – namely, the relative size of the two fields. Mostly everyone knows that graduate students in the social sciences outnumber those in computer science; and yet, once provided with a description of Tom W. as a high schooler, participants quickly called up their stereotypes about computer scientists and made a decision on the basis of this easily accessible information, without retrieving the crucial base-rate information that should have been factored into their responses as well. This example illustrates the intuitive mind's tendency to make use of nothing other than the most salient and accessible information. In the case of the inherence heuristic, this information will often be about the inherent characteristics of the to-be-explained pattern's constituents.

Once the mental shotgun has completed its job, the information generated is handed over to the next stage of heuristic processing. To use another one of Kahneman's (2011) metaphors, this next stage is a *storyteller*, looking to arrange the information at its disposal into a coherent narrative whenever possible (see Fig. 2). Whenever such a narrative emerges out of the assortment of facts called up by the mental shotgun, it then percolates up to working memory in the form of an apparently effortless intuition. Of course, such an intuition only *appears* effortless. It is actually the product of vast amounts of rapid processing that implicit cognitive processes perform behind the scenes.

To reiterate, the pool of facts activated by the mental shotgun for the purpose of generating an explanation for a pattern may often be heavily biased toward the inherent characteristics of that pattern's constituents. As a result, when the storytelling part of the heuristic process takes over and attempts to make sense of the information at its disposal, it will have a rather limited number of options. That is, it will often be forced to construct a story that explains the existence of a pattern in terms of the inherent features of the entities within that pattern rather than in terms of factors external to it. However, the one-sided nature of the information delivered by the mental shotgun is not an impediment to the storytelling process. Quite the contrary – the less information is available, the easier it will be to fit it all into a coherent story (Kahneman 2011). In the case of girls wearing pink, such stories are easy to construct: For example, perhaps girls wear pink because this color is flower-like and delicate – a perfect match for girls' delicate features. Likewise, the case of orange juice being consumed for breakfast can easily be fit into a sensible narrative. As the self-styled "world's undisputed #1 expert on breakfast" speculates in a post on his website, perhaps "the odor of citrus" is "energizing, invigorating and refreshing" and thus "helps wake you up" at the time of day when you need it the most.[5] Again, the storytelling stage has settled on an explanatory story that accounts for the existence of a pattern (we drink OJ in the morning) in terms of the inherent characteristics of the entities in that pattern (it's because OJ has an energizing smell that we drink it in the morning; see Fig. 2). More generally, the outcome of this stage will often be an intuition that the pattern under consideration can be explained by *the inherent features of its constituents*. Even in cases where a specific story fails to coalesce at this point, people may nevertheless be left with a vague sense that the inherent features activated by the mental shotgun will ultimately be sufficient to explain the pattern under

consideration (e.g., some of the typical characteristics of OJ and/or breakfast will ultimately explain why we drink OJ for breakfast, although it remains to be determined which ones).

Considering the fragmentary nature of the information it typically relies on, this heuristic process falls short of any normative standard of inference. Omitting from consideration large chunks of evidence simply because they didn't come to mind right away is guaranteed to introduce bias into the inferential process. The intuitive mind, however, seems to operate by the principle of "what you see is all there is": Any information that is not activated by the mental shotgun is completely ignored for purposes of making a judgment (see Fig. 2; Kahneman 2011). Similar to a jury that issues a verdict after hearing only one side of the argument, intuitive processes such as the inherence heuristic are prone to jump to conclusions based on the scant evidence pulled up by the shotgun. This obliviousness to the possibility that there were relevant facts that didn't immediately come to mind is a defining feature of intuitive processing.[6]

But why would people endorse these heuristic intuitions, especially given the strong possibility of bias? Why would we allow these intuitive processes to hijack our judgments? The consensus answer in the reasoning literature – an answer endorsed by many social psychologists as well (e.g., Anderson et al. 1996; Chen & Chaiken 1999; Gilbert & Malone 1995; Trope & Gaunt 1999) – seems to be that the human mind prefers the path of least resistance. That is, we are often content to allow quick heuristic intuitions to have the final say in circumstances where finding an alternative answer would require controlled, effortful processing. Evans (2006), for example, provided evidence for a *satisficing* principle, which he described as a "fundamental bias" to accept the output of intuitive heuristics "unless there is good reason to give it up" (p. 379). Likewise, Stanovich (2011) concluded that members of our species are *cognitive misers* who have a "strong bias to default to the simplest cognitive mechanism" (p. 29).[7] Thus, given this pervasive tendency to conserve effort, it is not surprising that shallow intuitive heuristics end up shaping so much of our reasoning.

In broad brushstrokes, this is how the inherence heuristic is hypothesized to operate. We might also describe this heuristic process as one of *question substitution*: "When confronted with a difficult question people often answer an easier one instead, usually being unaware of the substitution" (Kahneman & Frederick 2002, p. 53; see also Kahneman 2011). In the case of the inherence heuristic, the difficult question "What explains this pattern?" is often unwittingly answered as if it were the easier question "Which inherent features explain this pattern?" Many other intuitive heuristics can be described as inadvertent substitutions of this sort: When asked to rank probabilities, for example, people end up ranking similarity to a stereotype instead (the representativeness heuristic [Tversky & Kahneman 1983]); when asked to evaluate their overall life satisfaction, people end up evaluating their current mood instead (the mood heuristic [Schwarz & Clore 1983]); and so on. Importantly, in none of these cases are people substituting one question for another purposely, as a deliberate strategy. The intuitive answer simply pops into mind and is often accepted without much suspicion or scrutiny. The same is likely to hold true in the case of the inherence heuristic.

We now go on to address a number of more specific questions about the workings of this heuristic. Among other things, we clarify the notion of an inherent feature; we highlight additional inferences licensed by the output of the heuristic; and we discuss whether the intuitions to which the inherence heuristic gives rise can be blocked or revised, as well as whether there are developmental differences in the extent of its influence.

2.2.1. What counts as an inherent feature? We are claiming that the cognitive process that underlies the inherence heuristic (depicted in Fig. 2) often leads to explanatory intuitions couched in terms of inherent features. Here, we provide a brief clarification regarding our use of the term *inherent*. Roughly, our account classifies features as inherent if they can be said to characterize how an entity is constituted. Thus, inherent features tend to be stable, enduring characteristics of the entities in question – characteristics of the sort that might often be associated in semantic memory with those entities (e.g., OJ has a tangy taste; pink is delicate) and thus accessible during the shotgun phase of the heuristic. Note that this is a fairly permissive definition in that it encompasses a wide range of possibilities: Inherent features can be either concrete (e.g., OJ has vitamin C) or abstract and theory-laden (e.g., OJ is healthy), either superficial (e.g., men have facial hair) or deep (e.g., men have a Y chromosome), and so on. Even though this category of inherent factors is broad, there is of course much that falls outside its scope. For example, any aspects that are external to, or removed from, the focal entity do not count as inherent, even if they are strongly associated with it (e.g., OJ comes in half-gallon cartons; OJ is kept in the refrigerator). And, importantly, also falling outside this category are any considerations involving the history of the entity, prior circumstances that affected it, and so on. For example, the fact that pink used to be worn by boys would not qualify as an inherent feature, and neither would the fact that OJ was not consumed on a wide scale before the 1920s.

2.2.2. Patterns explained via the heuristic are seen as stable and inevitable. The intuition that observed patterns are explained by the inherent features of their constituents may license additional inferences about these patterns. Of particular interest to us in this section are inferences about the presumed stability and inevitability of patterns explained via the inherence heuristic: If a pattern is understood as being rooted in the very nature of the things that make it up, then it becomes difficult to imagine how this pattern could be otherwise. For example, the intuition that girls wear pink because pink is an inherently delicate color seems, on the face of it, to preclude the possibility that there used to be a time when boys wore pink, or the possibility that pink has begun to dominate girls' clothing only in the very recent past. In other words, the inherence heuristic is likely to cast the girls/pink mapping as a permanent, unavoidable fixture of human life. This perspective contrasts with that promoted by explanations in terms of extrinsic–historical factors (e.g., past events). These explanatory factors specify a starting point to the existence of the relevant patterns, revealing their temporally restricted nature, and highlight the possibility that things could have turned out differently, revealing the contingent nature of these patterns.

2.2.3. Variability in the explanatory "stories" used by the heuristic and thus in its output. As should be clear by now, the output of the inherence heuristic takes diverse forms. Although inherent features figure prominently in this output, the explanatory glue that binds these features together will vary quite a bit. One reason for this variability is that there are often multiple stories that can weave a plausible narrative out of the information pulled up by the mental shotgun. Which of these stories is actually incorporated into a particular heuristic judgment depends on factors such as the context in which the judgment is made and the prior knowledge that is brought to bear on it. Context matters because it can prime certain explanatory frameworks, making them temporarily more accessible to intuitive processing (e.g., Devine 1989; Higgins et al. 1977). For instance, watching an episode of *Grey's Anatomy* might increase the accessibility of biological explanations, as might having a stomachache. Similarly, context can temporarily highlight certain inherent features from among the pool of candidates, thereby influencing who the main protagonists are in the storytelling process (e.g., the features of girls vs. the features of the color pink). Context works on a broader temporal scale as well. For example, one's cultural setting can make certain explanatory constructs *chronically* accessible to heuristic processes, thereby regularly skewing the output of the heuristic in a particular direction. A person's prior knowledge may likewise boost the chronic accessibility of certain classes of explanations. The more extensive one's prior knowledge and use of a construct is, the more frequently this construct may be co-opted on subsequent occasions for the purpose of generating sensible stories (e.g., Bargh et al. 1986; Higgins et al. 1982).

The intuitions generated by the inherence heuristic vary not just within a pattern but *across* patterns as well. The source of this other type of variability is simply that the "stories" that are sensible for one observed pattern are often inadequate for another pattern. Although it may make sense, for instance, to rely on one's theories about disease prevention to explain why OJ is a morning drink (perhaps a dose of vitamin C early in the day is just what our immune systems need to ward off disease), these theories are of little use when considering why girls wear pink or why giraffes are called *giraffes*. Different explanatory notions are relevant to different types of entities in the world, and this basic fact will undoubtedly influence the storytelling step of the inherence heuristic. Depending on the regularity under consideration, the output of the inherence heuristic might appeal to a range of causal frameworks (biological, psychological, physical, etc.), as well as to value-laden, normative notions such as appropriateness, naturalness, or optimality (see especially sects. 3.3 and 3.4). For example, even though one might explain the pairing of orange juice with breakfast in terms of a specific biological mechanism (e.g., the properties of OJ boost immunity for the upcoming day), the pairing of pink with girls might perhaps be better accounted for via an optimality-based story (e.g., the features of girls and pink are an ideal match). In sum, the inherence heuristic makes opportunistic use of a variety of causal and normative explanatory notions in its attempt to cobble together a believable story.

Importantly, despite the substantial variability in the structure of the intuitions generated by the inherence heuristic, they will typically have a crucial aspect in common: They will, more often than not, explain an observed pattern as a function of its constituents' inherent features. This is the hallmark of the proposed heuristic process.

2.2.4. Scope of the heuristic: Patterns versus instances. So far, we have discussed the inherence heuristic as if it pertained exclusively to how people make sense of large-scale patterns (e.g., why so many girls wear pink). This is probably an oversimplification. In all likelihood, attempts to explain general patterns and specific instances (that is, specific events, outcomes, or behaviors) trigger the same intuitive process: The mental shotgun quickly generates some facts; these facts get passed on to the storytelling stage for further processing; and so on. Thus, more comprehensively conceived, the inherence heuristic is the intuitive process that is invoked to explain a *wide range of observations* (both general and specific) and that – due to shortcuts built into its structure – leads to an overreliance on inherence-based explanations.

If the same process is called up for both patterns and instances, then what justifies the focus on patterns? The answer is that the output of this common process may often diverge for the two types of inputs. Although there are good reasons to suspect that people overweigh inherent factors even in the case of single observations (e.g., Gilbert & Malone 1995; Jones & Davis 1965; Jones & Harris 1967), this intuitive process may be *particularly* likely to generate inherence-based explanations when it targets broad patterns, as we argue next. Thus, because the departures from normativity are most dramatic in the case of patterns, we have framed the inherence heuristic as targeting primarily this class of inputs. We will continue with this framing for the rest of the article as well (except for sect. 5.3, where we discuss the correspondence bias), on the understanding that it is a simplification.

Why might inherent explanations be more common for patterns than for instances? One reason for expecting such a difference is that information about applicable extrinsic factors may be somewhat more accessible in the case of specific observations (e.g., Cimpian & Cadena 2010; Cimpian & Erickson 2012; Cimpian & Markman 2009; 2011; see also Kelley 1967; 1973). During our everyday interactions with the world, we witness mostly specific events involving specific entities (objects, people, etc.). As a result, we have extensive opportunities to discover and record in memory the extrinsic causal factors that are relevant to such events (e.g., one's wireless router might stop working after it was accidentally stepped on). Arguably, these opportunities for discovery are less plentiful in the case of broad patterns, especially if these patterns consist of instances widely dispersed over space and time. Moreover, many of the extrinsic factors that are applicable to single events would – if called up to explain broad patterns as well – be hard to fit into a story that makes sense (e.g., accidents are not frequent enough to account for the high rate of interruptions in the wireless signal). As a consequence, our explanatory intuitions may end up relying particularly heavily on inherent factors (e.g., wireless routers are unreliable) in the case of patterns.

2.2.5. Blocking and revising the typical output of the heuristic. Although inherence-based intuitions exert a powerful influence on our understanding, their hold is

not inescapable. We are in fact capable of avoiding, as well as discarding, the inherence-based intuitions typically supplied by the heuristic, even if we may not do so on a regular basis. For example, on the select occasions when the mental shotgun successfully activates information about plausible extrinsic factors, this information may prevent these typical inherence-based inferences from being generated in the first place – that is, these inferences may be *blocked*. Alternatively, if an inherence-based explanation has already been generated, exposure to evidence inconsistent with it may lead to a *revision* of this explanation. Although many observed patterns may be understood initially as due to the very constitution of the things that make up these patterns, the alternative to this view is often within our grasp.

2.2.5.1. Individual differences. The alternative to inherent thinking may well be within our grasp, but not everyone is equally likely to reach out for this alternative. In other words, there may be individual differences in people's ability and motivation to overcome the tendency to adopt inherence-based explanations. Extensive evidence for such individual differences has accumulated with respect to intuitive heuristics more generally (e.g., Stanovich 1999; 2011; Stanovich & West 1997; 2000; see also Evans 2003; 2008; Kahneman & Frederick 2002). Extrapolating from this evidence, two dimensions of individual variability may be particularly predictive of the extent to which people rely on the typical (inherence-based) output of the inherence heuristic. The first dimension is *cognitive ability*, as measured by IQ and other similar tests (e.g., Stanovich 1999; 2011). Individuals with higher cognitive abilities possess more efficient systems for retrieving information from long-term memory, as well as more efficient mechanisms for processing the information retrieved (e.g., Cattell 1986; Lohman 1989). As a result, they may be more likely to activate (at the shotgun stage) and use (at the storytelling stage) information other than the highly available inherent features, which may in turn enable these individuals to avoid the usual output of the inherence heuristic.

Another aspect of general cognitive ability that might be relevant here is *cognitive control* – that is, the ability to regulate one's mental processes (e.g., Kane & Engle 2002; Stanovich 2011, Ch. 3). With greater cognitive control, high-ability individuals may be able to limit the influence of the "what you see is all there is" principle – that is, they may be able to avoid jumping to conclusions too quickly, based on too limited a pool of facts. Alternatively, if the inherence heuristic has run its course and an intuition has been generated, individuals with greater cognitive control may also be in a better position to scrutinize, rather than simply go along with, this heuristic intuition. That is, these individuals may be less likely to *satisfice* (Evans 2006) and thus more likely to resist the initial suggestions made by the inherence heuristic.

A second dimension of individual variability that may be predictive of reliance on the typical output of the inherence heuristic is *cognitive style* (e.g., Baron 2005; Evans 2008; Stanovich 1999; 2011). Individuals differ not only in their *ability* to perform complex cognitive computations (as already mentioned) but also in their *inclination* to perform such computations: Some people seek out tasks that challenge their abilities, whereas others gravitate toward familiar routines; some people relish gray areas, whereas others welcome the certainty of black-and-white opinions; and so on. The more favorably one is disposed toward engaged, open-minded thinking, the less one is susceptible to the influence of intuitive heuristics – even after accounting for cognitive ability per se (e.g., Stanovich 1999; 2011). We expect the same conclusion to hold with respect to the proposed inherence heuristic. Cognitive style could affect the operation of the heuristic at multiple points: for example, by modulating how thorough the information searches are during the shotgun stage, or by influencing how wary people are about going along with "what you see is all there is" (and thus how likely they are to jump to facile conclusions). Some evidence for the claim that open-minded cognitive styles reduce the influence of the inherence heuristic, as well as for the hypothesized negative relationship between cognitive ability and reliance on the inherence heuristic, is presented in section 4.3.

2.2.5.2. Developmental differences. The inherence heuristic is a basic cognitive process that operates in largely the same way across development. Nevertheless, the influence of this process may be more extensive in early childhood, narrowing somewhat as children become increasingly able to block and revise the inherence-based intuitions typically generated by the heuristic. Evidence for this claim (namely, that children are particularly prone to understand the patterns of their world via the inherence heuristic) will be discussed throughout the article (see also Cimpian & Steinberg, in press). It is not the case, however, that adults' reasoning is free from the influence of the heuristic. They too reason inherently, and quite often. Our claim here is simply that adults may be in a somewhat better position than children to overcome this way of thinking, as we explain next.

Inherence-based intuitions may be more prevalent in children's thinking for several reasons. First, children's causal-historical knowledge of the world is sparser than that of adults, which means that the facts available to children for purposes of weaving a plausible story are even more biased toward typical characteristics than are the facts usually available to adults. However, as children accumulate more information about the external factors that are relevant to the origins of observed regularities (e.g., historical events, social conventions), they may become increasingly likely either to block the typical output of the heuristic or to revise their original intuitions about an inherent basis for these regularities. For example, although many children might first explain why fire trucks are red by appealing to the inherent features of fire trucks and/or the color red, exposure to evidence of systematic variability in the color of fire trucks (e.g., they are yellow in Hawaii) might lead children to discard this inherent notion in favor of an explanation based on arbitrary social conventions.

Second, developmental differences in reliance on the inherence heuristic may also occur because of developmental differences in cognitive resources. Over the course of childhood, basic cognitive operations, including inhibitory control, undergo tremendous development (e.g., Kail 1991; Williams et al. 1999). Improvements in the infrastructure of cognition are likely to have noticeable effects on the proposed heuristic process by enabling children to perform faster and more comprehensive memory searches, for example, or by enabling them to resist going with the first thought that comes to mind. With age, then, children

may be better able to rein in the influence of intuitive judgments (e.g., Kokis et al. 2002).

However, it is also important to point out that development may not always bring about a reduction in the prevalence of inherence-based intuitions. The process that underlies the inherence heuristic is subject to influence from multiple factors, and some of these factors may offset the contributions of greater knowledge and enhanced cognitive efficiency. For example, adults may be more *motivated* to reach certain inherent conclusions than children are, especially when these conclusions produce desirable psychological effects (e.g., alleviating anxiety; see sect. 3.2). In these cases, motives would steer the course of the heuristic back toward inherence-based intuitions, canceling out the influence of improved cognitive resources. Because of interactions such as these, the trajectory of inherent thinking across development may not always be a simple downward trend.

2.2.6. Not always wrong. The inherence heuristic does not invariably lead to error. If certain aspects of the world cluster together reliably (e.g., people wear heavy clothing in the winter), one possibility is that they do so because of something internal to the cluster itself – because some elements of this cluster explain the presence of the other elements (e.g., the winter cold explains the use of heavier clothing). In other words, the inherence heuristic is capable of generating normatively correct judgments, which is also true of intuitive heuristics as a class (e.g., Evans 2009; Stanovich 2011). We should reiterate, however, that the inherence heuristic provides merely an *approximate* solution to the problem of explaining observed patterns. Inherent features are only a subset of the explanatory factors that could be at play, and ignoring the possibility that extrinsic (e.g., historical, social-conventional) factors may also be involved is undoubtedly a source of bias.[8]

2.3. Interim summary

The inherence heuristic is a fast, implicit cognitive process that often leads people to explain observed patterns in terms of the inherent features of the entities that constitute these patterns. In turn, these explanations instill a certain perspective on the patterns that are being explained, making them seem stable and inevitable. Although inherent features are typically present in the output of the heuristic, the explanatory structures that bind them together are likely to vary. The storytelling step of the inherence heuristic calls on a broad spectrum of beliefs about causal mechanisms (e.g., biological, psychological), as well as on normative, value-laden notions (e.g., optimality, appropriateness). As a result, there is a fair amount of diversity in the form of the intuitions generated by the heuristic, as subsequent sections will illustrate. The ultimate fate of the intuitions typically resulting from the inherence heuristic is, in principle, also varied – inherence-based intuitions are not obligatorily endorsed. Rather, they can be blocked before they are even generated (e.g., if information about plausible extrinsic–historical factors is activated by the mental shotgun); they can be scrutinized and rejected once they are generated; and they can be revised after they have been endorsed. However, under most ordinary circumstances, we expect that the output of the heuristic will be accepted as is and will thus influence much of how people understand the world around them.

3. The inherence heuristic as a unified explanation for diverse psychological phenomena

We proposed that people tend to make sense of observed patterns in terms of the inherent features of their constituents. Our goal in this section is to provide evidence for this proposal. Most of the evidence reported here was not originally collected to test our claims, so we will thus reinterpret it from our viewpoint. However, by positing the presence of an inherence heuristic, we will be able to highlight and explain the deep similarities across a broad range of findings that may at first appear unrelated. In and of itself, the fact that the inherence heuristic can provide a unified account for such a diversity of psychological phenomena highlights its worth as a scientific hypothesis.

3.1. Evidence from directly elicited explanations

Perhaps not surprisingly, several illustrations of the inherence heuristic can be found in research that directly elicited people's explanations for patterns. We begin with an example in which the heuristic leads to reasonably accurate (perhaps even normatively correct) explanations, followed by other examples that make more obvious the bias that the inherence heuristic can introduce into our judgments.

In one series of studies (Kelley 1967; 1973; McArthur 1972; Orvis et al. 1975), college students were asked to explain an event (e.g., why Amy laughed at Beth's joke) in the context of some additional information. Crucially, this additional information suggested either that the event in question was part of a broader pattern or that it was a one-off occurrence. Consistent with our proposal, participants were significantly more likely to invoke inherent features (e.g., aspects of Amy or Beth's personalities) when a pattern was present than when it was not. For example, if the information suggested that many other people laugh at Beth's jokes (a pattern), then the typical explanation was that Amy laughed at Beth's joke because of something inherent about Beth (e.g., she is a funny person) (Orvis et al. 1975). Similarly, if the information suggested that Amy laughs at pretty much any joke (a pattern), then the typical explanation was that Amy laughed at Beth's joke because of something inherent about Amy (e.g., she is a kind person or perhaps easily amused). In contrast, if the information suggested that the event to be explained was unique (e.g., nobody else laughed at that joke, and Amy has never laughed at Beth's jokes before), then the more common explanation appealed to "something about the particular circumstances at the time" (p. 608).[9] (To reiterate an earlier point, people may not fully escape the pull of inherent explanations even when reasoning about isolated behaviors, as suggested by the correspondence bias in person perception [e.g., Jones & Harris 1967]. However, Kelley and colleagues' evidence does seem to suggest that inherent explanations may be less prevalent for one-off behaviors.)

In simplified versions of these studies, even 4-year-old children appealed to inherent features to explain patterns of behavior (e.g., DiVitto & McArthur 1978; Higgins & Bryant 1982; Ruble et al. 1979; Seiver et al. 2013), which

is consistent with the claim that the inherence heuristic is available early in development. Evidence for inherence-based reasoning in children may also be found in the research on praise and criticism, as we explain next. Providing feedback for a specific outcome via a statement that generalizes beyond that outcome, and thus implies the existence of a pattern, often prompts the recipients of this feedback to draw conclusions about inherent qualities such as talent or aptitude (e.g., Cimpian et al. 2007; Kamins & Dweck 1999; Mueller & Dweck 1998). In turn, these inferences can undermine motivation and performance in later situations because they lead to anxiety about the level of these supposed inherent qualities (e.g., "How much talent do I have?") and about how failure might reflect on them. To illustrate, 4-year-old children who were praised for their success on a drawing task with the general praise "You are a good drawer" reacted more negatively to subsequent mistakes (e.g., displayed more negative emotions, were quicker to give up) than children who were given more specific praise, such as "You did a good job drawing" (Cimpian et al. 2007; see also Cimpian 2010; Cimpian et al. 2012). Presumably, this difference emerged because children's explanations diverged in the two cases – that is, because the praise that framed children's success as part of a broader pattern of successes was more likely to promote explanations in terms of inherent qualities than the specific praise was. This interpretation is admittedly one step removed from the data, since children were not actually asked to identify the source of their performance. Nevertheless, it seems quite likely that diverging explanations were at the root of children's behavior, especially in light of the evidence that beliefs about fixed talents (also known as "entity" theories [see Dweck 1999; 2006]) often give rise to the sort of helpless behavior that was observed after the general praise.

The conclusion that children explain regularities in inherent terms receives additional support from research on explanations for novel facts about categories and individuals (Cimpian & Cadena 2010; Cimpian & Erickson 2012; Cimpian & Markman 2009; 2011). Across a number of studies, 4- and 5-year-old children provided significantly more inherence-based explanations for the category-wide facts than for the individual-specific facts, although these two sets of facts were otherwise perfectly matched in content. For example, unfamiliar abilities that were said to characterize an entire social group (e.g., "boys are really good at a game called *gorp*") were often inferred to be due to inherent traits (e.g., "boys are smart") (Cimpian & Erickson 2012; Cimpian & Markman 2011). In contrast, children were significantly less likely to rely on inherent features when the same unfamiliar abilities were said to characterize a single individual (e.g., "there's a boy who is really good at a game called *gorp*"); children explained the abilities of individuals more often in terms of the historical process by which these abilities arose (e.g., "he practiced a lot").[10] Although such a process would probably be responsible for the abilities of a group as well (e.g., Romanians excel at gymnastics because of their rigorous training, not because they are naturally limber), children ignored this possibility and concluded instead that these patterns were due to the inherent features of their constituents. Thus, children's responses reveal the bias that the inherence heuristic often introduces into the process of generating explanations. Finally, these results also highlight how broadly the inherence heuristic applies, skewing people's interpretation of novel, as well as familiar, patterns. Even though children had no specific knowledge about gorp or any other activity we asked them about, their explanations nevertheless gravitated toward the (presumed) inherent properties of these activities and of the social groups said to be good at them.

The evidence reviewed in this section provides support for the claim that people make sense of broad facts about the world via an inherence heuristic. In the next section, we turn to a different literature and argue that the inherence heuristic may be part of the reason why people tend to see the current state of affairs in their society as fair and legitimate.

3.2. Evidence from research on system justification tendencies

Many people, including many who are disenfranchised, believe that the sociopolitical systems they are part of are "good, fair, natural, desirable, and even inevitable" (Jost et al. 2004, p. 887; see also Jost & Banaji 1994; Jost & Hunyady 2005; Kay & Friesen 2011; Kay et al. 2009; Lerner 1980; Liviatan & Jost 2011; Samuelson & Zeckhauser 1988). What is at the root of this striking belief? Considerable evidence suggests that the main culprit here may be a tendency to explain current societal arrangements as being due to the inherent characteristics of the people who make up the various strata of the social hierarchy (e.g., Gabennesch 1990; Hoffman & Hurst 1990; Jost 2001; Jost & Burgess 2000; Napier et al. 2006). For example, people who are poor must be so because they lack some of the traits needed to succeed in life (e.g., they must not be very hardworking, motivated, or intelligent) and not as a result of adverse circumstances or unfair biases in the opportunity structure of one's society. Or, to take another example, the fact that men hold political power must be due to the "fact" that they possess the intellectual and personality characteristics needed to lead (e.g., they are rational and level-headed) rather than to historical events that happened to favor patriarchal societies. This tendency to justify existing societal patterns by casting them as the inevitable products of inherent features is quite robust insofar as it is present even when people reason about unfamiliar circumstances. In one series of studies, for example, participants were told about two groups of aliens "who inhabit a distant planet" and who fulfill different roles in that planet's society (e.g., Orinthians tend to work "in the free-enterprise sector," whereas Ackmians tend to work "in the research/educational sector") (Hoffman & Hurst 1990). As predicted by the inherence heuristic account, participants thought that this pattern was in place because of the inherent qualities of the aliens in the two groups, and more specifically because of the perfect match between the aliens' characteristics and the stereotypical requirements of their respective occupations (e.g., Orinthians must be more self-assured and sociable, whereas Ackmians must be more intellectual and solitary).

These system-legitimizing inferences fit well with our proposal, in that they reveal a tendency to explain current societal patterns by appeal to the inherent features of the constituents of these patterns. There is, however, a dimension to these inferences that goes beyond what our account

can explain in its current form. Specifically, people are *motivated* to adopt system-legitimizing beliefs of this sort in order to alleviate discomfort or anxiety about their place in society and in order to achieve a sense of meaning, order, and predictability in their lives (e.g., Jost & Hunyady 2005; Kay & Friesen 2011; Kay et al. 2009). If current societal structures are fair and legitimate, then one's place in them is exactly where it should be, and thus there is no need to feel frustrated (if one is disadvantaged) or guilty (if one is privileged).

According to our proposal, people's use of the inherence heuristic is not driven by the need to reduce negative affect. Rather, this heuristic is a cognitive shortcut that operates seamlessly behind the scenes, shaping our explanatory intuitions about observed patterns. It is possible, however, that motives could exert a subtle influence at several points in the heuristic process. For example, the motive to defend the status quo might lead people to keep certain information (such as negative stereotypes of disadvantaged groups) active in memory and thus easily accessible to a variety of cognitive processes (e.g., Anderson et al. 1996; Higgins 1996; Higgins & King 1981). The increased accessibility of this information might then make it more likely to be retrieved by the mental shotgun and therefore more likely to be used during the subsequent storytelling process (see Fig. 2), thereby tilting the output of the heuristic toward intuitions that legitimize the political status quo. System justification motives may also influence how people handle the output of this heuristic process. When the inherence heuristic gives rise to intuitions that satisfy defense motives, people may be even more likely than usual to satisfice – to fail to scrutinize the output of the heuristic – and perhaps also more eager to protect this output from subsequent revision (e.g., Anderson et al. 1996; Chen & Chaiken 1999; Kruglanski 1996; Pyszczynski & Greenberg 1987). In light of these considerations, the evidence that system-justifying beliefs have a motivated component does not negate the possibility that these beliefs stem from the operation of the inherence heuristic. It is still quite possible that the proposed heuristic serves as the cognitive bedrock on which people can build a motivated system-justifying ideology.

Our claim that system justification tendencies are made possible by the inherence heuristic casts this well-established phenomenon in a new light. If we are correct, system-justifying intuitions are a subset of the inherence-based intuitions that people generate to make sense of all sorts of observed patterns, many of which have nothing to do with their social or political status. Although inherence-based intuitions that legitimize the status quo may be particularly common (in part because of their palliative effects), as well as particularly consequential for how we relate to one another, the process by which they are generated is no different than the process that leads people to conclude, say, that orange juice is consumed for breakfast because of its inherent properties (e.g., its energizing smell, its vitamin C content). By identifying the cognitive underpinnings of system-justifying ideologies, this provocative hypothesis also highlights the deep similarities between this and other important social-cognitive phenomena such the correspondence bias (sect. 5.3) and essentialism (sect. 4), both of which may be rooted in the same heuristic process.

Finally, this new perspective on system justification can be used to derive surprising new predictions. For instance, robust system-justifying tendencies should be found even in children's thinking, especially since the inherence heuristic may hold more sway at earlier ages (sect. 2.2.5.2; for some preliminary support for this prediction, see Baron & Banaji 2009). These inferences wouldn't serve the same palliative function for children as they do for adults (because children aren't typically as concerned about their place in society), but they should nevertheless be present early in development if they are in fact an output of the inherence heuristic. We might also predict that endorsement of system-justifying intuitions should be affected by the same variables that are hypothesized to affect endorsement of inherence-based intuitions more generally (e.g., cognitive ability, executive functioning, cognitive style). More broadly, our proposal might be extended to provide new insights into the cognitive foundations of political attitudes. For example, given that political conservatism is associated with both system-justifying tendencies and ambiguity-intolerant cognitive styles (e.g., Jost et al. 2003; Matthews et al. 2009), we might predict that, relative to liberals, conservatives would show greater reliance on the intuitions supplied by the inherence heuristic, even outside the realm of politics. To the extent that predictions such as these are supported by future work, they would further demonstrate the value of understanding system-justifying intuitions as emerging from the operation of the inherence heuristic.

3.3. Evidence from research on nominal realism

Consistent with the proposed tendency to explain observed patterns as being rooted in the characteristics of their constituents, children all over the world appear to believe that words are, to some extent, inherently compatible with the objects to which they refer. This belief is known as *nominal realism*. Of course, the vocabulary of a language is in reality the product of a long chain of historical events rather than a reflection of the inherent features of the things named. There is, for example, nothing sun-like about the English word *sun*. The pattern whereby speakers of English use this word to refer to the bright object in the sky is an arbitrary convention that arose over time to coordinate communication about that object.[11] And yet, our understanding of words as mere conventions is surprisingly fragile, especially in childhood. Piaget (1929/1967), for example, asked children whether it would have been possible for things in the world (e.g., the sun) to have different names than they currently do (e.g., *moon*). Children as old as 9 denied this possibility, citing reasons such as "Because the sun can't change, it can't become smaller" or "Because it's nothing else but the sun, it couldn't have another name" (p. 81). These responses suggest that, even for children well into their school years, names are not conventional patterns imposed on the world for the purpose of coordinating communication. Rather, names are believed to match the features of their referents – they are believed to "come from the things themselves" (p. 86). Gelman (2003) encountered the same belief in a conversation with a 9-year-old who was convinced that our names for dinosaur species could not be any different than they currently are because "they found the fossils and stuff" (p. 183) – again, as if names were inherent in the things named. Moving beyond the anecdotal, consistent evidence of childhood nominal realism was found in experimental research as well (e.g., Brook 1970;

Lockhart et al. 1977; Osherson & Markman 1975; but see Rosenblum & Pinker 1983).

Claims of nominal realism, however, might seem at odds with the developmental evidence suggesting that children understand the conventionality of linguistic forms (for a recent review, see Diesendruck & Markson 2011). This conflict is only apparent. What the evidence in this literature shows is that children expect words to be widely known to people in their linguistic community (e.g., Clark 1988; Diesendruck & Markson 2001; Graham et al. 2006). However, an expectation that words are widely known is quite different from, and does not entail, a belief that words are arbitrarily linked with their referents. After all, children expect many *nonarbitrary* facts about the world to be widely known as well (Cimpian & Scott 2012). It is thus entirely possible that children simultaneously (a) believe that words are conventional in the sense of *widely known* and (b) do not believe that words are conventional in the sense of *arbitrarily chosen*.

To return to our argument, it may not be just children who see words and objects as inherently, rather than arbitrarily, linked. Under certain circumstances, remnants of this belief can be identified in adults as well. A first piece of evidence for this idea is found in anthropological studies of nonliterate societies. Horton (1967), for example, claimed that people in these societies tend to "see a unique and intimate link between words and things" (p. 159; see also Frazer 1890/1958; Murdock et al. 1978). For many of them, words seem to be "inextricably bound up with reality" (p. 159), even to the point where a word might be used to stand in for the object it refers to (as in magic rituals, for instance). Assuming that Horton's claims are valid (cf. Scribner & Cole 1981; Tambiah 1968), the intuitions he describes here have all the telltale signs of the inherence heuristic. As people attempt to make sense of observed linguistic patterns (e.g., the pairing of *sun* with the sun), they might often arrive at the conclusion that these patterns are particularly appropriate: Perhaps, say, the sound of the word *sun* matches the features of its referent (e.g., its heat, its brightness) at some level. If people inferred such a match between the inherent features of a name and those of its referent, it would not be unreasonable for them to make the further assumption that an object's name could be used to stand in for the object itself (just as a photograph might be used to stand in for the object in it).

Traces of the belief in an inherent link between words and objects lurk beneath the surface in literate societies as well (see, for example, Rozin et al. 1986; 1990). As an illustration, we briefly describe the results of some of our own research on this question (Sutherland & Cimpian, in preparation). In these studies, our goals were to determine (a) whether we could in fact find traces of nominal realism in a sample of American adults and, if so, (b) whether individual differences in nominal realism would correlate with individual differences in more general inherence-based reasoning, as the present account might predict. To assess nominal realism, we created a scale that probed participants' ideas about the nature of word–referent links. For example, one item asked subjects whether "there is something particularly appropriate" about the name *giraffe*, or whether we "could have just as easily called this animal something else." Participants indicated their answers on Likert-type scales. (Responses to the item above, for example, were made on a scale from 1="This name is particularly appropriate" to 7="We could have easily called this animal something else.") We also asked participants to justify their answers to these scale items. Next, to assess broader inherence-based reasoning, we created an inherence heuristic scale that required participants to rate their agreement with statements such as "It seems natural that engagement rings typically have diamonds," "It seems right that black is the color associated with funerals," or "There are good reasons why orange juice is typically consumed for breakfast."[12] None of these items were about names or language, so the two scales were nonoverlapping in content.

The results of these studies were consistent with both of our predictions. First, traces of nominal realism were indeed detectable in our sample of American adults (all of whom were college students in this study). Consider, for example, the strikingly realist intuitions expressed in some of their open-ended justifications: One person explained why she thought the name *giraffe* was particularly appropriate by saying, "The animal has a long neck, so it is a *giraffe*"; another wrote, "I think the word *giraffe* captures the height well with the use of the double *f*s." Even when subjects answered that another name would have also been suitable, which was the more typical response, their justifications occasionally revealed an undertone of realism, as when a subject wrote that "we could have called it something that goes along with its features." The implicit assumption behind these answers seems to be that a name (even at the level of orthography, as with the double *f*s) is rooted in the inherent features of its referent.

Our second prediction was also supported: Participants' nominal realism scores were significantly correlated with their scores on the inherence heuristic scale, $r(124) = 0.31$, $p < 0.001$. This result is consistent with the argument that nominal realism is a specific instantiation of a more general propensity to explain the patterns of reality in terms of inherent factors. Moreover, the correlation between participants' nominal realism and inherence heuristic scores remained significant, $r(122) = 0.30$, $p = 0.001$, even when we partialed out the variance attributable to two cognitive variables that could have conceivably influenced both of these scores: the need for certainty and orderliness (known as *need for closure* [Kruglanski et al. 1993]) and the desire to engage in effortful cognitive activity (known as *need for cognition* [Cacioppo et al. 1984]). (The link between inherent reasoning and nominal realism was replicated in two additional studies in which we asked participants about artifact names [e.g., *bottle*] instead of animal names [e.g., *giraffe*] and included a number of other control measures.)

Interestingly, the inherence-based intuitions identified in our studies seem to extend to idiomatic expressions as well. That is, people often have the intuition that idioms (e.g., *taking the bull by the horns*) are transparently rooted in the features of their referents (Keysar & Bly 1995; 1999). To many of us, for example, the idiomatic expression *taking the bull by the horns* seems uniquely suited to describe the responsible act of assuming control of a difficult situation. In reality, however, the mapping between form and meaning in idioms such as these is far more arbitrary than our (arguably inherent or realist) intuitions would lead us to believe. *Taking the bull by the horns* could have just as easily meant the opposite of what it does today – that

is, it could have referred to the irresponsible act of taking unnecessary risks such as grabbing a raging bull by its horns. The fact that we use it to describe a responsible rather than an irresponsible act is probably nothing more than an accident of history – an accident that we nevertheless understand in inherent terms. Thus, for patterns of idiom use just as much as for patterns of word use, people tend to overestimate the extent to which language matches the inherent features of the world.[13]

To summarize, the phenomena described in this section support the argument that people tend to explain observed patterns by appealing to the inherent features of the things that instantiate these patterns. More specifically, people seem prone to explain word–object mappings in terms of the match between the features of the words and the features of the objects to which they refer. As a side note, the evidence reviewed in this section also illustrates the explanatory promiscuity of the inherence heuristic (in particular, its storytelling step), insofar as people's intuitive explanations for word–object links seem to rely on normative, rather than causal, considerations (see sect. 2.2.3). That is, the intuition that names are ideally suited for their referents because they provide an optimal match for their referents' features does its explanatory work via value-laden judgments about what is suitable and optimal.

3.4. Evidence from research on is–ought errors

As suggested in the preceding section, the intuitions due to the inherence heuristic are often normative: They explain via value judgments about what is appropriate, ideal, optimal, or right. In this section, we argue that – when used as explanations for patterns of human behavior – these heuristic intuitions are likely to introduce considerable bias into *moral reasoning*.

The sort of bias we have in mind was first identified by David Hume (1740/2000), who famously cautioned against placing undue weight on the patterns of current reality (i.e., what *is*) in drawing conclusions about moral truths (i.e., what *ought to be*). Imagine, for example, that you were a member of a social group whose members routinely hanged those who were suspected of theft. What factors should you consider in deciding whether such punishment is morally justified? According to Hume, the current prevalence of this punishment cannot be the only basis for your decision; additional premises are needed to justify conclusions about the ethics of an act other than its being typical (see also Sober 1994; Wilson et al. 2003). In the terminology of the inherence heuristic proposal, you would be in error to proceed directly from observing a pattern (e.g., hanging is a common punishment for theft) to inferring that this pattern is due to an optimal match between the features of its constituents (e.g., the severity of hanging matches the seriousness of theft-related offenses, and thus hanging is appropriate punishment for thieves).[14]

Despite Hume's centuries-old warning, is-to-ought transitions remain prevalent in our thinking, as would be expected if they were the output of an effortless heuristic shortcut. Indeed, empirical studies suggest that modern-day undergraduates are very much inclined to take *what is* as an unerring guide to *what ought to be* (Eidelman et al. 2009; Friedrich 2005; 2010; Friedrich et al. 1989). Across a number of studies, for example, participants rated currently observed patterns (e.g., a certain set of graduation requirements) as "good," "right," and "the way things ought to be" relative to hypothetical alternatives (Eidelman et al. 2009). Participants did so even though it was made clear that the change to these hypothetical alternatives could have some benefits and would occur at negligible cost. Here as well, then, we may see evidence for the heuristic intuition that the world is as it is, and should not be any different, because of the inherent features of the things in it: The current graduation requirements are in place because they are inherently appropriate, perhaps even optimal, and not because they are the outcome of a contingent historical process that could have just as easily followed a different path.

Consistent with the developmental predictions of our account (see sect. 2.2.5.2), children's ability to keep *is* separate from *ought* may be even more fragile than adults'. This is not to say that children imbue all social patterns with moral force: Young children are often able to differentiate among social regularities, realizing for example that some of these exist due to context-specific conventions (e.g., at our school, we sit in a circle during story time) whereas others are in place due to context-independent considerations of justice and welfare (e.g., we refrain from hitting others and stealing their toys) (for a recent review, see Helwig & Turiel 2011). However, even if children understand *some* conventional patterns as being conventional, there is also evidence of developmental change in this respect. That is, in comparison with older children and adults, younger children seem to still reify a good number of the patterns that structure their everyday lives, viewing these patterns as inherently appropriate rather than as dependent on social consensus (see Gabennesch 1990; Gelman & Kalish 1993; Kalish 1998; Kalish & Lawson 2008; Lockhart et al. 1977; Shweder et al. 1987; Smetana et al. 1993; but see Helwig et al. 1990).

To illustrate, Kalish (1998) presented preschoolers with stories whose protagonists were said to *want* to violate a social or physical regularity. For instance, one boy wanted to wear a dress (social violation), and another wanted to turn into a bird (physical violation). The participating children were then asked if the protagonists would actually perform these actions (e.g., "Will he really. . . ?"). In a separate study, children were also asked to make predictions about the actions of protagonists who not only wanted to violate these regularities but were also ignorant of them. For instance, the boy in the story wanted to wear a dress and, not being "from around here," didn't know that boys don't wear dresses. As we would predict, the younger preschoolers in these studies appeared to treat violations of social and physical regularities as equally improbable. That is, 3½-year-olds often denied that the characters in these stories would violate the relevant regularities, even when these concerned (what are in reality) social conventions, and even when the potential violator was both ignorant of these conventions and motivated to act counter to them (see also Lockhart et al. 1977). These responses are compatible with the claim that many social patterns hold considerable force for young children, as would be expected if they interpreted these patterns via the proposed heuristic. For instance, if the features of dresses make them inherently appropriate for girls, then of course any boy – even one who has never seen a dress before – would immediately recognize that wearing such a thing would be wrong.

By offering a hypothesis about the cognitive underpinnings of is–ought reasoning, our account opens up new possibilities for experimental investigation of this important phenomenon. To date, discussion of is–ought inferences has been confined mostly to philosophy, with relatively little attention paid to their characteristics as a psychological phenomenon (for a few exceptions, see Eidelman et al. 2009; Friedrich 2005; 2010; Friedrich et al. 1989). If this type of reasoning were indeed an output of the inherence heuristic process (see Fig. 2), then its likelihood of occurring should be sensitive to the same factors that modulate endorsement of inherence-based intuitions more generally (e.g., cognitive ability, cognitive style; see sect. 2.2.5). Moreover, is–ought reasoning should pattern with other phenomena that are thought to stem from this heuristic, such as nominal realism and system justification. The presence of structural commonalities between these phenomena might also suggest that conclusions drawn about one should extend to the others. For instance, it is possible that is–ought reasoning would be influenced by defense motives, just as system-justifying reasoning seems to be. Is–ought beliefs (whose core implication is that the patterns of our world are exactly how they should be) may be adopted more often by people with a chronic need for such comforting thoughts, as well as in situations that temporarily heighten this need. Importantly, the influence of motives should not be limited to is–ought inferences regarding one's place in society, which also fall under the purview of system justification; rather, this influence should be felt with respect to a much broader range of inferences about what is right or appropriate. As these considerations illustrate, the inherence heuristic proposal is a generative source of insights into the diverse phenomena that are hypothesized to stem from this heuristic.

3.5. Interim summary

We set out to provide evidence for our proposal that people often understand the patterns they detect in the world as being due to the inherent features of the entities that instantiate these patterns. We found evidence for this proposal in a diverse set of findings that – although not originally collected to test our account – were all parsimoniously interpretable in terms of a single underlying process: namely, the inherence heuristic. By revealing the shared underpinnings of well-established phenomena that have previously been regarded as distinct, our proposal also provides a fresh perspective on these phenomena and a rich basis for further empirical research.

4. The inherence heuristic as a precursor to essentialist thinking

A major goal of this article is to outline the possibility that the inherence heuristic lays the cognitive foundation for the emergence of psychological essentialism. Specifically, we will propose that the essentialist framework emerges as an elaboration of the earlier, and more inchoate, intuitions supplied by the inherence heuristic. Before detailing this proposal, however, we explain what psychological essentialism is, highlighting its importance both as a basic feature of the human mind and as a potential source of prejudice and conflict.

4.1. What is psychological essentialism?

Psychological essentialism is an influential hypothesis regarding the structure of natural and social kind concepts (e.g., Gelman 2003; Haslam et al. 2000; Medin & Ortony 1989; Newman & Keil 2008; Prentice & Miller 2007; Rhodes & Gelman 2009; see also Bloom 2000; 2004). According to this hypothesis, people routinely assume that different kinds of entities in the world (e.g., lions, tulips, gay people) are underlain by different unseen, but real and causally potent, *essences*.[15] In other words, people have consistent (yet somewhat vague) intuitions about the causal source of the features that characterize various kinds: a certain internal, physical, microstructural *je ne sais quoi* that is unique to each kind and that invariably causes its members to display the full complement of typical features (barring accidents). For example, people might reason that lions look the way they do (e.g., tawny) and behave the way they do (e.g., hunting gazelles) because they all possess lion DNA or something of the sort, which is also the reason they are lions rather than donkeys or frogs. Similarly, people might reason that water looks the way it does (e.g., clear) and behaves the way it does (e.g., freezing at 32°F) because of some special fact about its microscopic structure, which is also what makes it water rather than ethanol or granite. The presence of this implicit belief in essences is supported by an impressive range of findings and has been documented in children as young as age 4 (for a comprehensive review, see Gelman 2003).

On a more practical level, because essentialist beliefs imply that the differences between categories are deep and immutable, they are also a likely source of intergroup prejudice and hostility of the sort that is common in human societies (e.g., Dar-Nimrod & Heine 2011; Haslam & Whelan, 2008; Prentice & Miller 2007). For example, the belief that social groups have essences has been associated with stronger endorsement of stereotypes (e.g., Bastian & Haslam 2006; Levy et al. 1998; see also Yzerbyt et al. 2001), greater acceptance of racial inequities (Williams & Eberhardt 2008), and more blatant prejudice toward members of minority groups (e.g., Keller 2005; see also Haslam et al. 2002).

4.2. The origins of essentialism: Preliminary considerations

Given the centrality of essentialist beliefs to human cognition and their role in fostering prejudice, conflict, and inequality, it seems particularly important that we understand the origins of essentialism itself. How does the human tendency to essentialize arise? To start, the consensus among scientists seems to be that the categories essentialized by humans do not in actuality have essences (e.g., Dupré 1981; Leslie 2013; Maglo 2011; Needham 2000; 2011; Sober 1994). Thus, essentialist assumptions cannot come from the world per se, for the simple reason that our world is not carved at the joints by real essences. Nor are these assumptions merely a product of socialization. Detailed examinations of parent–child conversations have found little in the way of overt essentialist talk to children (Gelman et al. 1998; 2004). At best, parental input provides subtle, indirect hints (e.g., kind-referring generic nouns) that would not be sufficient to create an essentialist

mind-set from whole cloth (e.g., Cimpian 2013; Gelman et al. 2010; Rhodes et al. 2012). It seems most probable, then, that essentialism is an endogenous feature of the human mind: either an innate module that evolved as a product of natural selection (e.g., Atran 1993; 1998; Gil-White 2001; Pinker 1994, Ch. 13) or the result of a highly constrained developmental process whereby children construct essentialist beliefs out of simpler cognitive parts, many of which may themselves be innate (Gelman 2003; but see also Bloom 2000, Ch. 6; Carey 1996; Kornblith 1995, Ch. 5).

Because the latter appears to be the more promising of the two alternatives, as argued persuasively by Gelman (2003, Ch. 11), we adopt it here as a working hypothesis.[16] On this account, the essentialist framework is elaborated out of developmentally prior abilities and biases (e.g., the ability to distinguish surface appearances from underlying reality) over the first few years of life. These preexisting cognitive capacities are clearly not essentialism per se, but they each supply distinct functional content that is necessary to the development of a mature essentialist framework. To clarify, the claim here is not that essentialism is just a collection of separate processes, without any psychological reality of its own. Although the origins of essentialism lie in prior capacities, the essentialist framework transcends their sum.

4.3. The inherence heuristic as a precursor to essentialism

Our proposal is that *psychological essentialism emerges as an elaboration of the intuitions supplied by the inherence heuristic*. To reiterate, the inherence heuristic often gives rise to intuitions that explain observed patterns in terms of the inherent features of their constituents. Which inherent features play the starring role in these intuitions depends to a large extent on the storytelling step of the heuristic. If, for example, people make sense of the fact that orange juice is consumed for breakfast via an explanatory story about the rousing properties of citrus smells, then these smells will be highlighted as a key source of this pattern. Alternatively, if the story used is one about the immunity-boosting properties of vitamin C, then it will be the nutritional content of orange juice that is highlighted as a source of the pattern. Or again, if no plausible stories are identified at this step, the heuristic will output a more inchoate sense that some to-be-determined combination of inherent features is responsible in some to-be-determined fashion for the observed pattern (an output that may be refined at a later time if applicable explanations are identified; see sect. 2.2). The sparser the explanatory options available to the storytelling step of the heuristic are, the more often it will be forced to produce the vague, "to-be-determined" sort of output. Given that the store of explanatory options is more limited for children than for adults, the output of the inherence heuristic may be particularly open-ended in early childhood – and thus more often in need of further elaboration.

On our proposal, the early output of the inherence heuristic is gradually elaborated into a full-blown essentialist stance over the first few years of life. This early output is not *itself* essentialism. Unlike essentialist beliefs, for instance, inherence-based intuitions do not necessarily pinpoint a single inherent feature as the source of the multiple regularities observed within a kind. Similarly, the inherent features present in the early output of the heuristic do not necessarily fit the description of an essence – they may not be thought of as necessarily physical, internal, and nonobvious, like essences are. Despite such differences in content, however, these inherence-based intuitions make it possible for children to *eventually* formulate essentialist beliefs. In other words, although children may at first understand the patterns that characterize natural and social kinds via vague intuitions couched in terms of to-be-determined inherent factors, they may soon begin to fix some of the free parameters in this equation, coming to more specific conclusions about the nature of these inherent factors. The end result of such an elaboration process, insofar as it occurs, may be the belief in an essence. More precisely, this elaboration process may lead to the belief that a *single* inherent feature explains the regularities observed within a kind – most likely a *physical* feature that is *internal* to the members of this kind and thus *nonobvious*. For example, children may eventually come to believe that *lions* and *water* display the features they do because of some unique feature of their internal structures or, later in development, perhaps because of something about their DNA and chemical composition, respectively.

How does the transition to a more essence-like notion of inherent cause occur? Are there explanatory stories or beliefs that could guide this elaboration process, and are these beliefs developmentally prior to essentialism per se? We describe two possibilities for how this process might unfold, focusing in particular on essentialism about living kinds. (Essentialist beliefs about other domains are discussed later in this section.)

First, the initial output of the inherence heuristic may be elaborated on the basis of children's ideas about *insides*. Although young children know relatively few specifics about the inside contents of objects (e.g., Gottfried & Gelman 2005; Simons & Keil 1995), they do seem to possess an abstract expectation that insides are *causally powerful* (e.g., Gelman & Wellman 1991; Newman et al. 2008), and particularly so for animate kinds (Gelman 1990). For instance, 14-month-old infants expect novel animate objects with similar insides to exhibit similar patterns of movement and sound, even when their surface markings differ (Newman et al. 2008; see also Welder & Graham 2006). Thus, even before the onset of essentialism (Gelman 2003), children's reasoning about animate objects includes an abstract assumption that internal parts are causally responsible for these objects' behavior. Note that this early causal bias is not tantamount to a belief in a category essence: Initially, the insides are seen simply as an enabler of movement and behavior, not as the identity-conferring source of all properties that characterize the relevant category. Even so, such a skeletal causal bias might be sufficient to guide the initial output of the inherence heuristic toward the notion of an internal, nonobvious (i.e., essence-like) inherent cause. Whereas children might start out agnostic concerning which inherent features explain why the members of, say, the *lion* kind exhibit all the regularities they do (e.g., being tawny, having manes, hunting gazelles), the presence of an early bias to endow internal parts with causal powers might soon enable children to refine this view: Whatever it is that makes lions what they are, it probably has to do with their insides. Moreover, since lions display a set of regularities that is distinct from that of

any other kind, this internal *je ne sais quoi* that lions have must also be distinct from that of any other kind.[17]

Children's early reasoning about animate objects also features the construct of *internal energy*, which could provide an alternative means of elaborating the initial output of the inherence heuristic. That is, young children seem to assume that animate beings possess some sort of vital substance that is responsible for maintaining life (e.g., Hatano & Inagaki 1994; Morris et al. 2000) and for providing the causal force behind movement and growth (e.g., Gelman & Gottfried 1996; Gottfried & Gelman 2005). The roots of this causal belief may be present even in infancy. For example, 5- to 6-month-olds' reasoning about the behavior of self-propelled objects suggests that infants attribute to these objects a source of internal energy that makes it possible for them to behave independently of external physical forces and thus, say, resist movement when pushed or even float when released in midair without adequate support (Luo et al. 2009). Again, this early notion of an internal, nonobvious source of energy that is causally responsible for the observable behaviors of animate objects does not amount to a belief in category essences. Yet, this notion may still provide an adequate means of refining the inchoate early output of the inherence heuristic. It is clear, though, that the essentialist belief constructed via this refinement process must transcend children's original ideas about internal energy, both because the essence is *specific to a kind* rather than being a general causal force and because the essence is causally responsible not only for dynamic features such as movement and growth but for *static* features as well.

To review, at least two sorts of explanatory beliefs (beliefs about insides and beliefs about internal energy) could in principle guide children from an inchoate sense that the patterns encountered within a living kind are due to some combination of inherent factors to more concrete ideas about these factors. The result of this elaboration process, we propose, could function as the category essence. It is important to note, however, that on either of the alternative paths sketched above, the end point will not be terribly concrete in an absolute sense; it will simply be more concrete relative to the starting point. That is, we suspect that children typically narrow down the possibilities only to the level of a *type* of inherent factor: something physical, internal, nonobvious. It is rare even for adults to go beyond this level of (non)specificity in their thinking about essences, which is why essences are characterized more accurately as *essence placeholders* (e.g., Gelman 2003; Medin & Ortony 1989): Although people believe that essences exist, they often cannot say much about them, perhaps other than that they are physical, internal entities that are in principle knowable at a detailed level (e.g., by the appropriate experts). Thus, the elaboration process that we proposed to account for the origin of essentialism probably has as its terminus point an essence placeholder rather than a specific essence.

Although we have so far focused mostly on essentialism about living kinds, the present proposal can also account for an important characteristic of essentialist beliefs – namely, the substantial cross-domain variability in their content (e.g., Bloom 2000; Gelman 2003; Haslam et al. 2000; Newman & Keil 2008). Such variability is a natural by-product of the mechanism hypothesized here. If essentialist beliefs are constructed by elaborating the open-ended output of the inherence heuristic on the basis of plausible explanatory stories, and if these stories differ depending on the entities under consideration (e.g., lions vs. New Yorkers vs. corkscrews), it follows that the end product of this elaboration process (that is, essentialist beliefs) should differ as well. We flesh out this argument via several examples.

First, children seem to endorse a more localized view of the essence for animate than for inanimate natural kinds (e.g., lions vs. gold) (see Newman & Keil, 2008, Experiment 2). Such differences in essentialist beliefs are consistent with our elaboration account insofar as there are also differences in the explanatory stories available to children in the two domains. As evidence of such domain differences, consider children's beliefs about the causal powers of insides. These beliefs appear to pertain specifically to the domain of animates (e.g., Gelman 1990; Gelman & Wellman 1991), with no evidence that children believe inanimate natural kinds such as *gold* to also have discrete internal parts or organs with causal properties. Thus, if the hypothesized elaboration process relies on different explanatory beliefs in these two domains, this process will most likely lead to different assumptions about where the essence is located. With respect to animates, the notion of insides as a causal source may license the inference that the essence is found where the vital insides are rather than being distributed homogeneously throughout. Depending on the explanatory notions used with respect to inanimates, children may either remain agnostic about the location of the essence or else adopt a distributed view.

Second, variability in the explanatory stories used during the elaboration process may also lead to variability in the content of people's beliefs about *what* the essence is (not just *where* it is). Some evidence for such variability is found within the social domain (e.g., Haslam et al. 2000). Although certain social groups (e.g., men, African-Americans) are thought to share an underlying biological essence, essentialist beliefs about other groups (e.g., New Yorkers, hippies) do not include a biological component but nevertheless cast these groups as coherent and homogeneous. Speculatively, it is possible that explanatory stories involving biological factors are deemed more plausible for social groups that have distinguishing biological markers (e.g., men) than for groups that do not have such markers (e.g., New Yorkers). Put another way, if some of the regularities that set a group apart from others include observable biological features (e.g., facial hair, deep voices), then a story that appeals to underlying biological factors unique to that group will seem more plausible and may thus be used more often in the elaboration process. Although the regularities detected within the social groups that do not have such observable biological markers are also likely to be interpreted via the proposed heuristic, the inherent features assumed to be responsible for these regularities may remain unspecified or may be fleshed out in psychological, rather than biological, terms (e.g., the regularities in New Yorkers' behavior may be due to something about their inherent personality traits).

Third, variability in the explanatory stories available to the elaboration process may also explain whether essentialist beliefs about a given category emerge at all or whether this process takes the initial output of the inherence heuristic in an altogether different direction. To take a salient

example, artifact concepts do not exhibit the hallmarks of psychological essentialism – or at least not the canonical forms of it (Gelman 2003; but see Bloom 1996; 2000; Hood & Bloom 2008). That is, artifacts (e.g., coins) are not typically assumed to possess an internal, microstructural category essence that is the causal source of their features. And yet, the regularities pertaining to artifact categories are, we suspect, interpreted via the same heuristic process that is used to make sense of the regularities pertaining to natural and social kinds. Why, then, do children's inherence-based intuitions about artifacts follow a different course from their intuitions about living things and natural objects? On our account, there are two main reasons for this divergence. The first reason is that there are no artifact counterparts to the early causal notions about insides and internal energy, which pertain exclusively to animate beings. As a result, there are fewer opportunities (if any) in this domain for children to flesh out the early output of the heuristic into more specific beliefs about causally powerful, internal essences. The second reason for this divergence is that the early output of the inherence heuristic with respect to many artifact categories is often overridden rather than elaborated. Children may *initially* understand the regularities observed within many familiar artifact categories in inherent terms, much as they do the regularities observed within natural and social kinds. For instance, young children may believe that coins are round, that fire trucks are red, and so on, because of some to-be-determined combination of inherent factors (Cimpian & Steinberg, in press). Relatively early on, however, children also start to understand the crucial role that people (an *extrinsic* cause) play in the emergence of these regularities (e.g., Gelman & Bloom 2000; Gelman & Gottfried 1996; Rhodes & Gelman 2009). As evidence accumulates in favor of this explanatory story, the appeal of earlier inherence-based intuitions may diminish to the point where children may actually abandon them. (As argued before, the intuitions supplied by the inherence heuristic are open to later revision if information comes along that is contradictory to these initial intuitions.) In sum, we argue that essentialism does not take hold in the artifact domain in part because (a) there are no plausible candidates for fleshing out the initial, open-ended output of the inherent heuristic into beliefs about causal essences, and (b) the initial output of the inherence heuristic is often revised or overridden on the basis of children's early-developing appreciation for the extrinsic sources of artifact properties.

In the final part of this section, we lay out a few predictions that follow uniquely from the proposal that the inherence heuristic provides the foundation for psychological essentialism. To begin, if the inherence heuristic is a precursor to essentialism, it would have to be available to children before the first signs of essentialism emerge. The developmental evidence available so far (see sects. 3.1, 3.3, and 3.4) suggests that children as young as age 4 exhibit reasoning patterns that bear the signature of the inherence heuristic. However, since 4 is also the age when children show the first clear signs of essentialist thought (Gelman 2003), evidence is needed about the extent to which children younger than 4 rely on the inherence heuristic. In addition, to minimize the possibility that the explanations of these younger children would stem from essentialism per se, it seems best to explore their inherent reasoning with respect to domains that are not typically thought to fall under the scope of essentialism even when it does eventually emerge (e.g., artifacts). For example, do 3-year-olds think that fire trucks are red for inherent reasons and not because people decided at some point that they should be red? Do 3-year-olds think that coins have always been and will always be round, as would be expected if they understood this pattern in inherent terms (see sect. 2.2.2)? If inherence-based reasoning of this sort were present in 3-year-olds, and we predict it should be, then it could plausibly serve as a foundation for later essentialist thought.

Beyond simply being present early on, the inherence heuristic should play a direct role in the subsequent emergence of essentialism. One way of examining this link would be to test whether individual differences in inherent thinking at age 3 correlate with individual differences in later essentialist thinking (say, at age 4). If essentialism is elaborated out of the intuitions generated by the inherence heuristic, as we argue here, then the strength of children's inherence-based intuitions at earlier ages should be predictive of the strength of their essentialist beliefs at later ages (but not vice versa). In fact, a similar prediction could be made with respect to the levels of essentialism observed in *adults*. There is considerable variability in the extent to which adults endorse essentialist beliefs, particularly those about social groups (e.g., Bastian & Haslam 2006; Keller 2005; Williams & Eberhardt 2008). According to our account, the people who are particularly likely to endorse essentialist beliefs should also be the people who are particularly likely to go along with, rather than scrutinize, the output of the inherence heuristic.

This last prediction was explored in a recent study from our lab (Salomon & Cimpian 2014). To measure reliance on the inherence heuristic, we used the scale described earlier, in section 3.3. This scale required participants to rate their agreement with a series of statements such as "There are good reasons orange juice is typically consumed for breakfast" and "It seems natural to use red in a traffic light to mean *stop*." To measure endorsement of essentialist views, we used a fairly standard scale of the sort devised by Haslam et al. (2000). The scale required participants to rate the extent to which social groups such as girls or Catholics are cohesive, have genetically determined traits, are rooted in an underlying reality, and so on. We designed these two scales so that there was minimal overlap in content between them. This feature of the study afforded a strong test of our prediction: It might seem counterintuitive to expect that people's beliefs about, say, whether it is natural to use red in a traffic light to mean *stop* would predict their beliefs about, say, whether girls are a cohesive group whose traits are determined by their genes. We also asked participants to fill out nine additional measures that enabled us to control for potentially confounding cognitive, personality, and ideological dimensions. For instance, a critic could argue that high levels of both essentialist and inherence-based reasoning might simply co-occur in people who are low in cognitive ability or whose cognitive style leads them to avoid to ambiguity and seek simple, black-or-white answers instead. The nine control scales were divided into three sets of three scales each, with each set administered to approximately a third of the participants (*N*s for the three subsamples ranged from 105 to 112; total $N = 323$).

The results were in line with our prediction. Participants' reliance on the inherence heuristic strongly predicted their

endorsement of essentialist views about social groups, $r(321)=0.39$, $p < 0.001$. Moreover, participants' inherence heuristic scores remained a significant predictor of their essentialism scores even when the nine cognitive and personality dimensions used as controls were taken into account, $rs > 0.33$, $ps < 0.001$. The unique predictive power of our scale reveals an intimate link between inherent thinking and endorsement of essentialist beliefs.

These results also provide insight into the factors that moderate the influence of the inherence heuristic on people's judgments. In section 2.2.5.1, we predicted that the tendency to go along with the easy intuitions supplied by the inherence heuristic should depend both on cognitive *ability* and on cognitive *style*. That is, people with higher cognitive abilities should be less likely to rely on the heuristic, as should be people whose thinking is more engaged and open-minded. With respect to cognitive ability, we indeed found that higher scores on this dimension (as measured by an abbreviated Raven's Progressive Matrices test [Raven 1960]) were associated with lower scores on the inherence heuristic scale, $r(103) = -0.28$, $p = 0.005$. Our predictions were confirmed with respect to cognitive style as well, insofar as participants' inherence heuristic scores were (a) negatively correlated with their scores on the Need for Cognition Scale (Cacioppo et al. 1984), which assesses the motivation to engage in effortful cognitive activity, $r(104) = -0.36$, $p < 0.001$, and (b) positively correlated with their scores on the Need for Closure Scale (Kruglanski et al. 1993), which assesses the motivation to arrive at simple, unambiguous judgments, $r(104) = 0.18$, $p = 0.056$. These results support the claim that the inherence heuristic provides a quick and easy means of making sense of observed patterns – a first guess that may be more easily overcome by individuals with extra cognitive resources or more reflective thinking styles.

To review, our proposal that essentialism emerges as an elaboration of the output of the inherence heuristic leads to several novel predictions. These predictions, some of which are already receiving empirical support, are also unique to our account. As a point of comparison, consider the prominent view that essentialism is an innate cognitive module that (a) was originally favored by natural selection because it "allows people to perceive and predict many important properties that link together the members of a biological species" (Atran 1998, p. 550; see also Pinker 1994, Ch. 13) and that (b) was co-opted, later in our evolutionary history, for reasoning about social categories such as ethnic groups, which shared certain characteristics with biological species (Gil-White 2001). As currently described, the modularity account provides little reason to expect that young children would have a broad tendency to view the patterns of their world, regardless of what sorts of entities they involve, in inherent terms. It also provides little reason to expect that this tendency to reason inherently would precede, and predict, the emergence of essentialism about natural and social kinds (the putative targets of the innate essentialist module). In addition to facilitating new insights into the roots of essentialist thinking, the present proposal can account quite naturally for some of the established findings in the literature, such as the domain differences in the content of essentialist thought. This is, again, an area where existing proposals struggle. For example, people's tendency to essentialize nonliving natural kinds such as gold or social groups such as New Yorkers is difficult to accommodate within the framework of a modularity account because these sorts of categories are not species-like in any straightforward way and thus fail to meet the input conditions of the hypothesized essentialist module.

4.4. Interim summary

This section outlined a new perspective on the process that leads to the emergence of essentialist views. Briefly, we proposed that the essentialist framework is constructed as a means of elaborating the inchoate intuitions supplied by the inherence heuristic. Although it is beyond doubt that there is theoretical and empirical work left to be done, the argument laid out here nevertheless holds considerable promise because it spells out a detailed, testable mechanistic model for the development of essentialism.

5. Distinguishing the inherence heuristic from related proposals and processes

This section is intended to clarify the scope and contribution of our proposal. To do so, we first differentiate it from two other hypotheses (concerning so-called K-laws and formal explanations) that seek to explain some of the same phenomena. We then go on to discuss the relation between the inherence heuristic and the correspondence bias, a cognitive process that leads to an overreliance on inherent explanations of human behavior. As we argue below, the correspondence bias may be viewed as a specific instantiation of the inherence heuristic.

5.1. K-laws

Strevens (2000; 2001) argued that people endorse what he termed *K-laws* (where the *K* stands for *kind*). According to such K-laws, membership in a natural kind is causally responsible for the observable features displayed by the members of that kind. For example, a tiger has stripes because something about being a tiger caused it to have stripes.

To begin, there are notable differences in the content of K-law-based and inherence-based explanations, both in terms of the explanatory entities that figure in the two types of explanations and in terms of the explanatory relations they use. On the first point, inherent features always do the explanatory work in the output of the heuristic. In contrast, at least early in development, K-laws are neutral about what aspects of category membership are causally efficacious (and thus explanatory). That is, children "might have no opinion about what does the causing" or might even think that "it is just a brute fact about the world" that membership in a category gives rise to category-typical features (Strevens 2000, p. 154). Thus, the K-law account makes no commitment to features of any sort serving as explanations for observed patterns. There are also differences in the explanatory relations used by the two accounts. The inherence heuristic is opportunistic: It relies on a variety of explanatory relations, both causal and normative (e.g., about what is optimal or appropriate). In contrast, K-laws are by definition causal, so they are able to explain only via causal relations. In sum, the explanatory intuitions due to K-laws are likely to differ on

multiple dimensions from the output of the inherence heuristic.

Also note that our proposal spells out in detail how the inherence heuristic operates, including why it so often leads to explanatory intuitions that rely on inherent rather than extrinsic factors. In contrast, Strevens provides no account of the process by which children come to endorse K-laws. Thus, even if the two proposals were equally successful in accounting for the phenomena they were intended to explain (but see Ahn et al. 2001; Meyer et al. 2013), the greater degree of mechanistic specificity in the inherence heuristic account may make it more fruitful as a working hypothesis.

Finally, there is a difference in the scope of the two accounts. As should be clear given the range of evidence reviewed in this article, the inherence heuristic is invoked to explain much more than the observable features of natural kinds, which is the explicit scope of K-laws. Insofar as the inherence heuristic proposal is actually successful in accounting for this broader range of evidence, its superior coverage – along with considerations of parsimony – would recommend it over the narrower K-law proposal.

5.2. Formal explanations of k-properties

According to Prasada and Dillingham (2006; 2009), people represent the relation between a kind and some of its properties as a part–whole relation: That is, some properties of a kind – dubbed *k-properties* – are represented as an *aspect* of being that kind of thing (e.g., having stripes is an aspect of, or a part of, being a tiger). Importantly for our purposes here, Prasada and Dillingham further argued that the part-whole relations between k-properties and their kinds license so-called *formal* explanations – that is, explanations that account for the presence of a k-property simply by appealing to the kind itself (e.g., tigers have stripes because they are tigers).

Prasada and Dillingham's proposal diverges from ours in many of the same respects that Strevens's did. For instance, there are clear differences in the content of the explanations involved. Most obviously, the proposed heuristic does not generate formal explanations of the sort Prasada and Dillingham describe; rather, its explanations rely on inherent features. One circumstance in which the two types of explanations might appear to overlap is when the heuristic's storytelling stage fails to converge on a plausible story. Although the more inchoate intuitions that emerge on these occasions sound roughly similar to both K-laws and formal explanations (e.g., girls like pink because of something about girls), the resemblance is superficial. Even when the heuristic produces this sort of vague output, the underlying assumption still is that some to-be-determined features are accounting for the relevant pattern, not the kind itself. Another major difference in the content of the two types of explanations is that inherence-based explanations are often *causal*, whereas formal explanations cannot be causal, by definition (Prasada & Dillingham 2006; 2009). However, the part–whole relations hypothesized by Prasada and Dillingham do license *normative* expectations for k-properties (e.g., if having stripes is part of being a tiger, then a tiger should have stripes), so the possibility of generating normative content is a characteristic that the inherence heuristic and the formal explanation accounts have in common.

It is also important to point out the differences in the scope of the two accounts. The inherence heuristic can be invoked to explain pretty much any observed pattern, whereas Prasada and Dillingham's formal explanations concern only those features of a kind that are represented as aspects of being that kind (namely, k-properties). In addition, even though there may be a role for noncausal, formal explanations in everyday cognition, people's reasoning about the world seems to rely heavily on causal explanations from the youngest of ages (e.g., Ahn et al. 2001; Gelman 2003; Medin & Ortony 1989; Schulz 2012; Strevens 2000). Thus, the inherence heuristic proposal is better equipped to provide a successful account of how children and adults make sense of their natural and social environments.

5.3. Correspondence bias

The term *correspondence bias* refers to a well-documented tendency to explain other individuals' behaviors (e.g., she is fidgeting) in terms of corresponding dispositional traits (e.g., she is an anxious person) rather than situational pressures (e.g., it's finals week) (Gilbert & Malone 1995; Jones & Harris 1967; Ross et al. 1977; Trope & Gaunt 2000; for recent reviews, see Gawronski 2004; Trope & Gaunt 2007). The clear parallels between this phenomenon and the inherence heuristic emerge, we argue, because the two share an underlying mechanism: That is, the intuition that observed behavior is due to stable psychological dispositions may stem from the same heuristic process that leads people to conclude that observed patterns are due to the inherent features of their constituents. This claim is consistent with the more comprehensive definition of the inherence heuristic we advocated for earlier (sect. 2.2.4), on which the heuristic provides an intuitive means of explaining specific instances and general patterns alike. Although we have focused mostly on patterns so far, the correspondence bias reveals that people's judgments veer away from normative rationality – and toward inherence – even in the case of specific instances.

Why would this be? With respect to behavior, one key reason is that extrinsic factors such as situations or norms are seldom on display in the actual behavior to be explained (Gilbert & Malone 1995). Although people do realize at some level that external pressures shape behavior (e.g., Gawronski 2004), there is little in the behavior itself to remind observers of these pressures. Therefore, a fast, ballistic process such as the inherence heuristic may often overlook these distal forces that influence behavior without leaving any visible clues or reminders of their influence. Instead, the heuristic process may end up (over-) relying on information about the actors themselves, since the actors are more transparently associated with the behavior to be explained than the relevant external constraints are. In particular, the actors' psychological characteristics (e.g., anxiousness) provide the heuristic with plausible causal stories for observed behavior and may thus figure prominently in people's intuitions – especially when the mental shotgun fails to activate thoughts about distal constraints. From our perspective, then, it may be reasonable to view the correspondence bias as emerging from the use of the inherence heuristic to explain particular instances of human behavior. (In fact, others have also argued that the bias is a result of heuristic, rather than systematic, processing [e.g., Trope & Gaunt 2007].)

The inherence heuristic framework we proposed here can accommodate not only the basic existence of a correspondence bias but also some of the nuances of this phenomenon. We list a few relevant findings and briefly point to the elements of our model that might account for them. First, the correspondence bias is more pronounced when participants' cognitive resources are taxed (e.g., Gilbert et al. 1988). As discussed in section 2.2.5.1, limitations in cognitive resources are also known to lead to greater reliance on intuitive heuristics (e.g., De Neys 2006; Stanovich & West 1997; 2000). Second, the correspondence bias is reduced when thoughts of situational constraints are made more accessible, either temporarily via priming (e.g., Trope & Gaunt 2000) or chronically via exposure to cultural contexts that emphasize the power of context in shaping human behavior (e.g., Morris & Peng 1994; but see Krull et al. 1999). On our proposal as well, the outcome of the heuristic's storytelling step is determined in part by the relative accessibility of the explanatory notions at its disposal (sect. 2.2.3). Factors that activate, either temporarily or chronically, thoughts of external causes should lead to a proportional decrease in inherence-based outputs. Third, the magnitude of the correspondence bias is modulated by motivational states. For example, observers who are assigned the goal of assessing the *situation* do not overuse dispositional explanations of behavior (e.g., Krull 1993). This conclusion is compatible with our inherence heuristic proposal, insofar as goals and motives can exert considerable influence over the course taken by the heuristic process (sect. 3.2).

These three examples, while not doing full justice to the richness of the literature on the correspondence bias, nevertheless highlight the plausibility of understanding this bias as a specific instantiation of a broader inherence heuristic. Looking to the future, this perspective on the correspondence bias also creates opportunities for cross-fertilization. In particular, the theoretical and methodological insights that have accumulated over 50 years of research on this bias in person perception could be used to guide new research on phenomena such as essentialism and system justification, which are relative newcomers to the psychology scene.

5.4. Interim summary

Our goal in this section was to clarify the relationships between the inherence heuristic and other accounts (K-laws and formal explanations) as well as other psychological tendencies (the correspondence bias) that might be thought to give rise to some of the same intuitions. In clarifying these relationships, we have also highlighted the unique theoretical contribution of the inherence heuristic account.

6. Conclusion

We proposed that human reasoning relies on an inherence heuristic, an implicit cognitive process that often leads people to explain observed patterns in terms of the inherent features of their constituents. This heuristic was argued to give rise to a diverse collection of intuitions and beliefs. Consider, for instance, the common belief that the political status quo is fair and natural. From the perspective of our account, the prevalence of this belief is due, at least in part, to the fact that people make sense of observed societal patterns (e.g., income disparities between groups) in terms of the inherent characteristics of the people that instantiate these patterns (e.g., their intelligence, their motivation; see sect. 3.2 on system justification). Or, to bring up another earlier example, the heuristic tendency to explain observed patterns in inherent terms may also be at the root of the linguistic intuition that words provide a suitable match to the features of their referents (see sect. 3.3 on nominal realism). Along the same lines, we argued that the inherence heuristic is a key reason why people reify observed patterns, imbuing them with moral force (see sect. 3.4 on is–ought errors). Arguments such as these led us to the conclusion that the inherence heuristic is a powerful and pervasive cognitive process. Additionally, these arguments brought to light the deep commonalities among important phenomena that had previously been considered unrelated. By uncovering such commonalities, we were able to offer new insights into the sources of these phenomena and to generate new directions for future research. Some of these directions are already being pursued in our lab, and promising results were summarized at several points in the article to underscore the generative nature of our account.

The other major goal of this article, aside from introducing the inherence heuristic, was to propose that the heuristic serves as a precursor to psychological essentialism. This belief in hidden essences is a central feature of human concepts of natural and social kinds – a feature that has also been linked to phenomena such as stereotyping, prejudice, and intergroup conflict. Building on the arguments in the first part of the article, we hypothesized that the origin of this important aspect of human psychology can be traced back to children's reliance on the inherence heuristic. That is, essentialism was proposed to emerge as an elaboration of earlier, and less specific, intuitions generated by the inherence heuristic. The hypothesized link between the inherence heuristic and the subsequent development of psychological essentialism was able to account for well-established findings (e.g., domain differences in the content of essentialist beliefs), and it led to several unique predictions as well (e.g., relationships between essentialism proper and more general inherent thinking about other aspects of the world). In sum, our proposal spelled out a plausible mechanistic model of the development of essentialism, a model that has the additional advantage of revealing the connections between essentialism and other inherence-based phenomena (e.g., system justification, correspondence bias).

Together, the arguments and evidence in this article help uncover a fundamental new aspect of human psychology. The inherence heuristic is a powerful cognitive process that provides an easy, but also bias-prone, means of explaining why the world is as it is.

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NOTES

1. The inherence heuristic operates only over the patterns or regularities that our minds represent as such, having either detected their presence in the world or inferred their existence via inductive generalizations (e.g., Gelman & Markman 1986; Markman 1989). The further question of which patterns are extracted from the world and which are not is complex and falls outside the scope of the current proposal.

2. For readers who are familiar with the dual-process literature, we intend the distinction between *deliberate* and *intuitive* processes to map roughly onto the distinction between System 2 and System 1 (e.g., Stanovich 1999) or type 2 and type 1 processes (e.g., Evans 2008). We chose not to use these other terms in the article both because their meaning is intuitively opaque and because they have misleading implications. (In particular, System 1 and System 2 are not really systems per se but rather heterogeneous collections of processes with loosely similar properties [e.g., Evans 2008; Stanovich 2011].)

3. This initial component has close counterparts in other theories of intuitive processing, although it usually goes by other, less colorful, names. For example, Stanovich (2011) describes a *fundamental computational bias* to "contextualize a problem with as much prior knowledge as is easily accessible" (p. 44; see also Stanovich 1999, Ch. 7). Similarly, Evans's (2006) *relevance principle* refers to "the powerful tendency to contextualize all problems with reference to prior knowledge elicited by contextual cues and the current goals that are being pursued" (p. 380).

4. Kahneman (2011) uses the term *mental shotgun* to refer more generally to the intuitive mind's tendency to perform a broad range of quick computations, often many more than needed, in response to a question or task. Our usage preserves some of this meaning but is also tailored to the specific process of the inherence heuristic.

5. http://www.mrbreakfast.com/ask.asp?askid=26 (accessed May 19, 2013).

6. Because intuitive heuristics use just the information that happened to be retrieved and ignore other, potentially contradictory, information that might nevertheless be available upon more careful reflection, they have been argued to introduce an element of *confirmation bias* into human judgment (Evans 2006; Kahneman 2011; for work on the confirmation bias, see Nickerson 1998; Snyder & Swann 1978; Wason 1960). Importantly, however, these heuristics do not reduce to the confirmation bias. For example, the inherence heuristic supplies explanations, whereas the confirmation bias does not – rather, the bias (in the form of the "what you see is all there is" principle) simply influences what sorts of evidence the heuristic process considers when formulating explanations.

7. The term *cognitive miser* has had a long tradition in social psychology as well (Fiske & Taylor 1984), although social psychologists typically use the term in a slightly different sense – as a strategic, deliberate choice to employ the simplest possible cognitive mechanisms (e.g., Bargh 1999; Gilovich & Griffin 2002) rather than as a fundamental (but defeasible) bias to default to such mechanisms.

8. The distinction between inherent and extrinsic explanatory factors is not always clear-cut. Some social conventions, for example, might themselves have an inherent basis (e.g., sneezing into the crook of one's arm is in fact an effective way to prevent the spread of germs). The existence of such ambiguous cases, however, does not negate the possibility that people routinely overestimate the importance of inherent explanatory factors.

9. Our interpretation of these responses as stemming from the operation of a fast heuristic diverges from the theoretical model that motivated Kelley et al.'s studies (Kelley 1967; 1973; Kelley & Michela 1980). According to Kelley's model, the explanation process is analogous to the formal procedures for hypothesis testing and data analysis used by empirical scientists and is thus likely to be slow, effortful, and deliberate (see Fiske & Taylor 1984). However, there is no evidence that participants in the studies reported here employed such effortful analytic strategies. Thus, Kelley et al.'s evidence might also be accounted for by the operation of a fast intuitive heuristic that leads people to the same conclusions via a different process.

10. Given the research on general vs. specific praise (e.g., Cimpian et al. 2007), we speculate that inherent explanations would have been even more infrequent in these studies if we had asked children to explain a single outcome involving an individual (e.g., "there's a boy who did really well…") instead of a regularity about an individual (e.g., "there's a boy who is really good…").

11. Despite some traces of nonarbitrariness in word–object mappings (such as onomatopoeia [e.g., *cuckoo*] or the *bouba/kiki* phenomenon [Maurer et al. 2006; Ramachandran & Hubbard 2001]), the vast majority of words do not in fact resemble or mimic the features of their referents in any meaningful way.

12. The inherence heuristic scale consisted of 15 items. In a sample of 323 participants, this scale displayed satisfactory internal consistency (Cronbach's alpha=0.85, mean item–total correlation=0.48). In addition, a confirmatory factor analysis provided support for the hypothesis that it measures a unidimensional construct – namely, reliance on the output of the inherence heuristic.

13. An argument could be made that people also understand *grammatical gender* in inherent terms (Boroditsky et al. 2003). In reality, there is no inherent reason why certain objects have names of a certain grammatical gender. For example, it's an arbitrary, contingent fact that the name for the sun is feminine in German (*die Sonne*) – after all, the sun is masculine in Spanish (*el sol*). Nevertheless, people often seem to assume that the grammatical gender of a noun matches the inherent features of the noun's referent (e.g., that *Sonne* is feminine because there is something feminine about the sun).

14. We point interested readers to Roberto Unger's (e.g., 1983) critique of the modern legal system, a critique whose main thrust is similar to Hume's. Unger argues that, although laws often do little more than codify the current realities of one's society, people (including lawyers and law scholars) often view them as stemming from transcendent (moral, political, etc.) principles. From a psychological perspective, this tendency to "sanctify the actual" (p. 571) by believing that laws reflect an absolute reality rather than the contingencies of one's time may be another instance of the inherence heuristic at work.

15. To be clear, psychological essentialism is a claim about the mind, not about the world. It posits that people believe in the existence of essences, not that essences actually exist in the world.

16. Briefly, Gelman's (2003) argument rests on two types of considerations. First, she argues that current versions of the modularity proposal cannot explain the full range of evidence regarding the development and scope of essentialist reasoning. Second, she argues that, given the existence of cognitive capacities that are developmentally prior and functionally related to essentialism, any proposal that suggests essentialism is an entirely separate module that does not draw on these existing capacities seems unparsimonious.

17. The question might also arise of whether the elaboration process proposed here leads to expectations about a single essence like inherent cause or multiple such causes for each category. Assuming that children make use of the same, most relevant and accessible, explanatory story to refine many of their inherence-based intuitions within a category, it is likely that they will also end up pinpointing the same cause (e.g., internal parts). The resulting single-cause structure is, in fact, typical of essentialist reasoning (e.g., Ahn et al. 2001).

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